

Classification, Localization , Detection

Object Classification

- Single Object
- Define to which class this object belongs to

Object Localization and Classification

- Single Object
- Define to which class this object belongs to
- Draw a bounding Box around the object location

Object Detection

- Multiple Objects
- Draw a bounding Box around the objects locations

Examples of Objects Classification, Localization, and Detection

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Person



Car

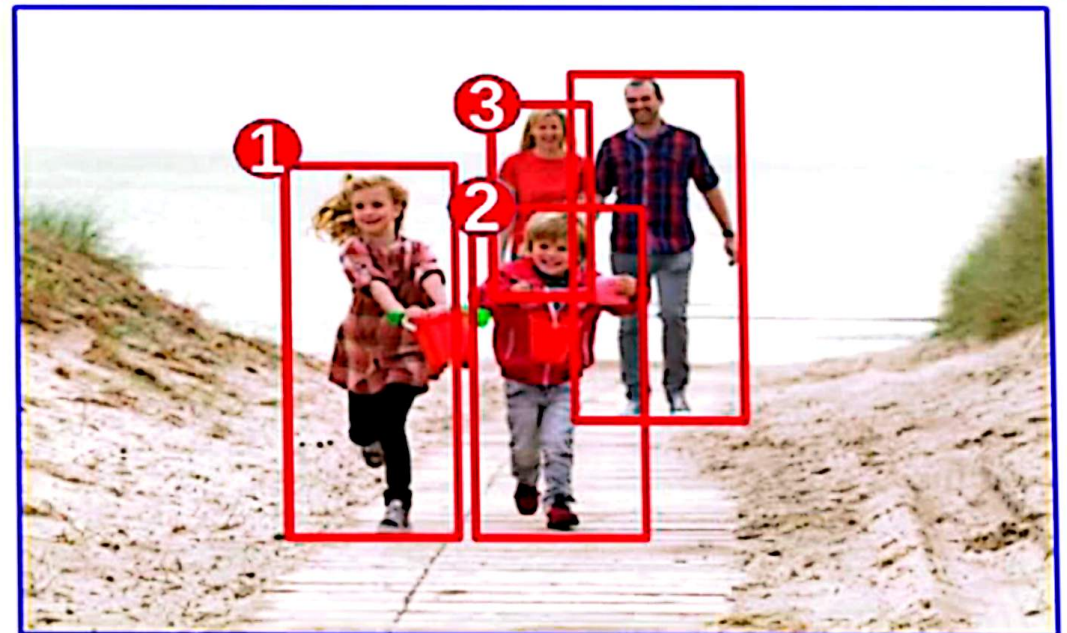


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Apple



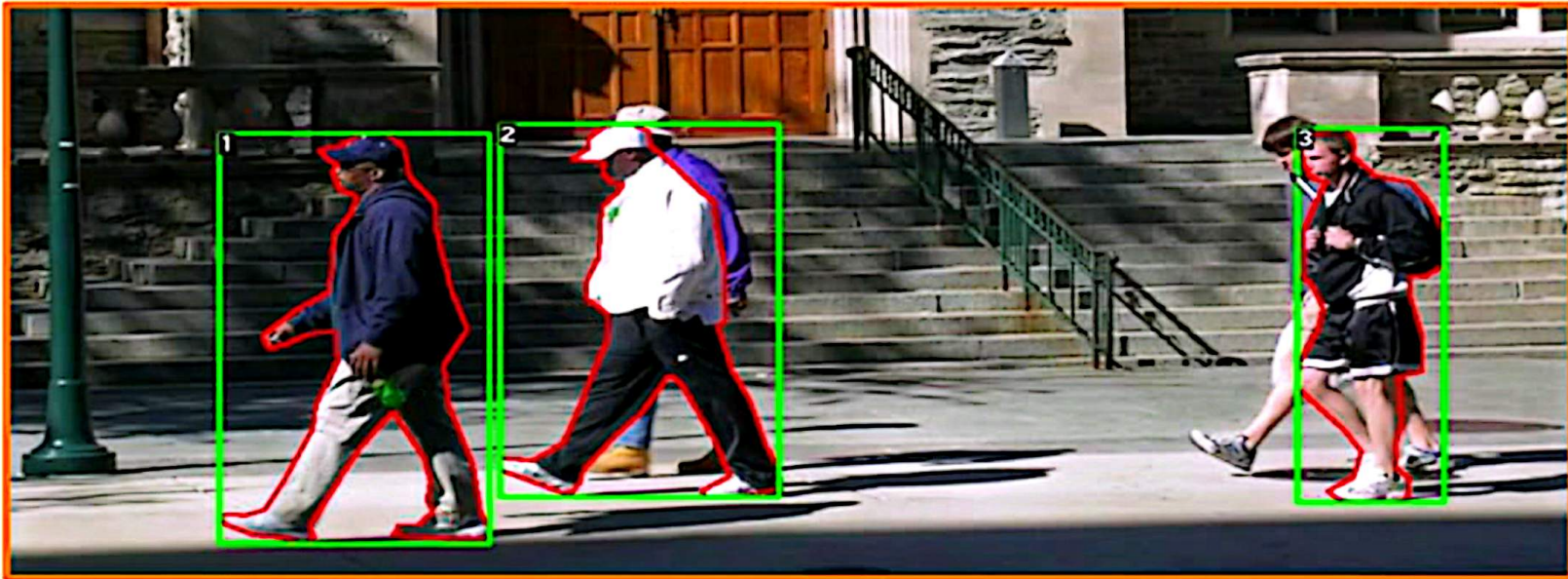
Person
X,Y,W,H



Detect Learned Objects ONLY

Localization vs Segmentation

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[1] Classification

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Document has just been saved...

[2] Localization

(0,0)



(1,1)

(0,0)

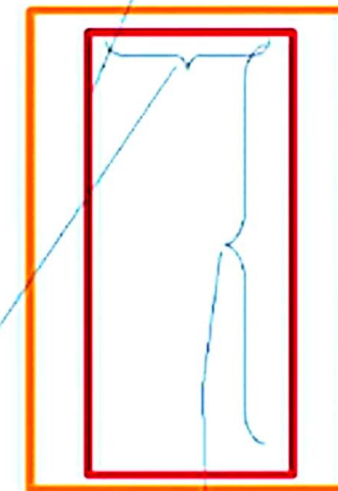


(1,1)

Normalized Image Width and Height (1,1)

(0,0)

$B_x=0.15$
 $B_y=0.1$



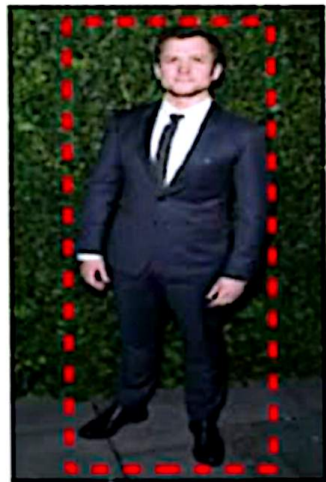
(1,1)

Width
 $W=0.5$

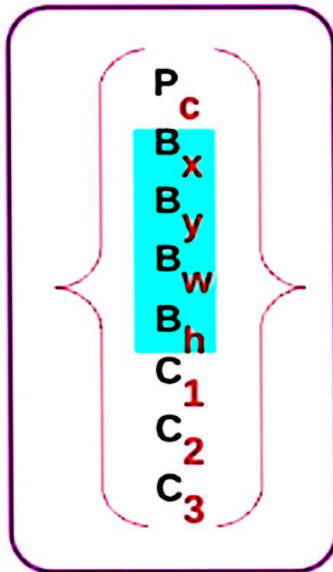
Height
 $H=0.9$

Localization and Classification

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Output
Y



Soft max Layer

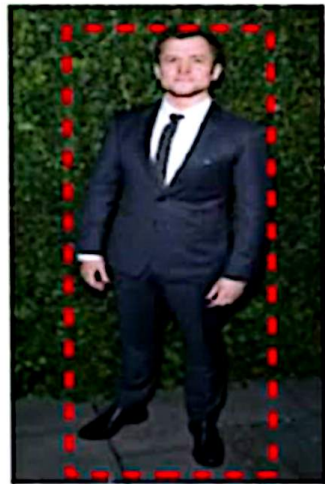


P_c is the Probability that there exist one of the Classes

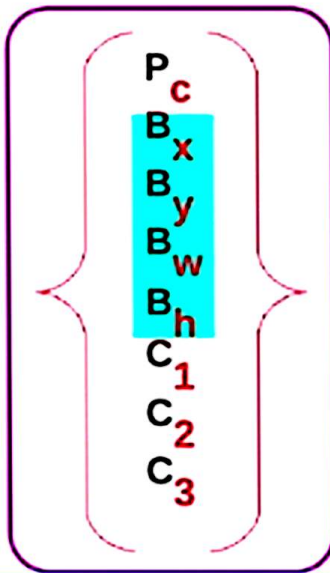
IF $P_c = 0 \Rightarrow$ All Values are Don't Cate

Localization and Classification

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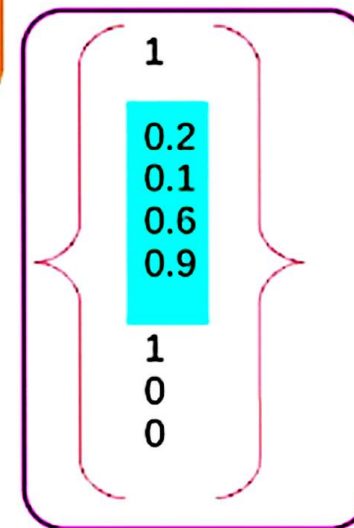
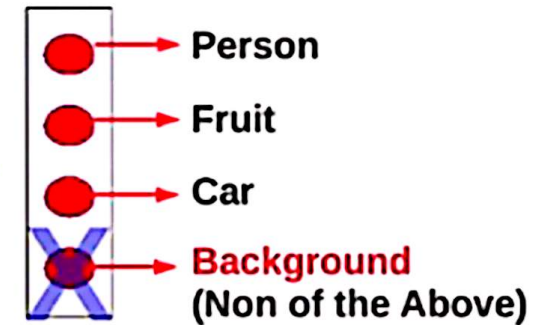


Output
Y



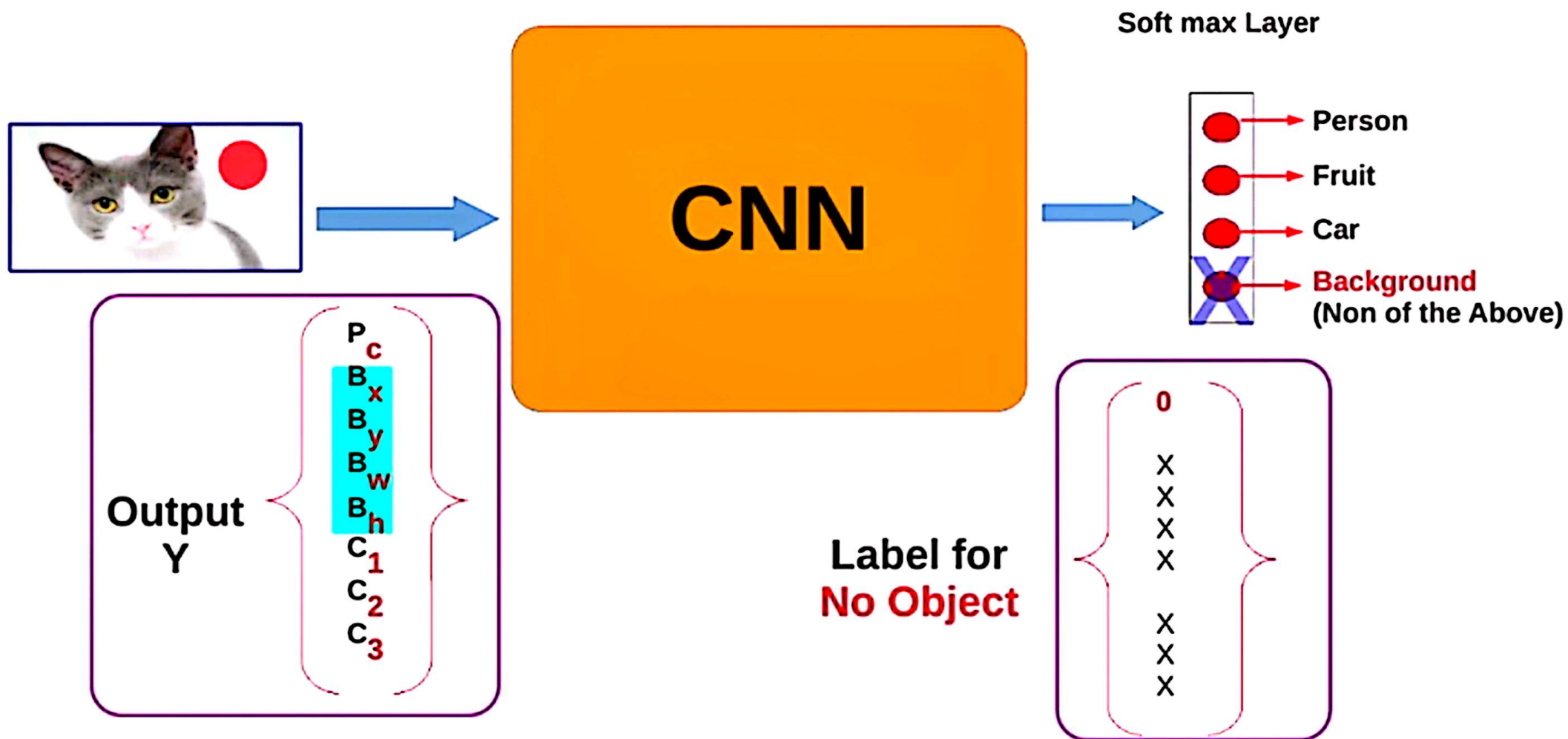
Label for
Person

Soft max Layer



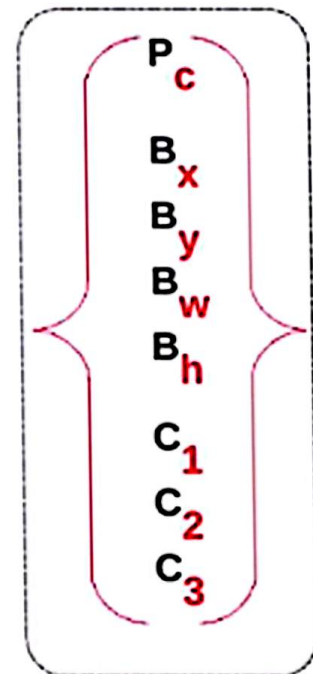
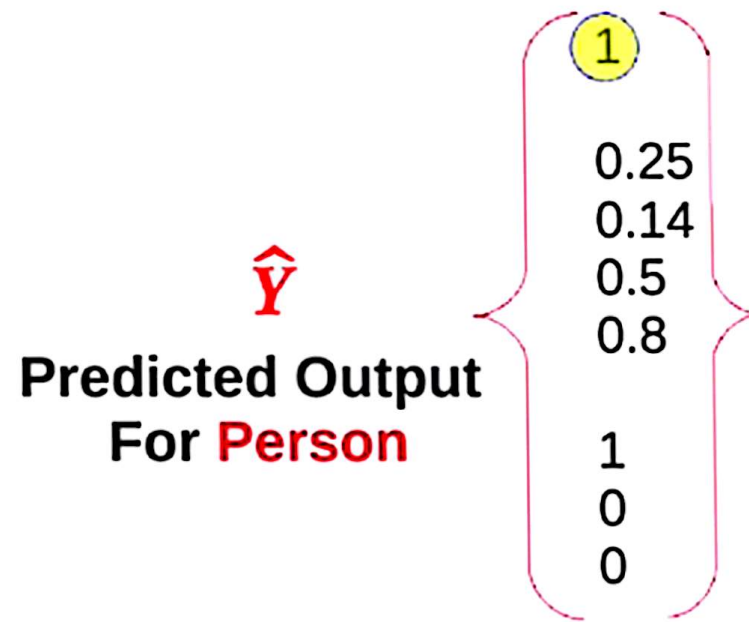
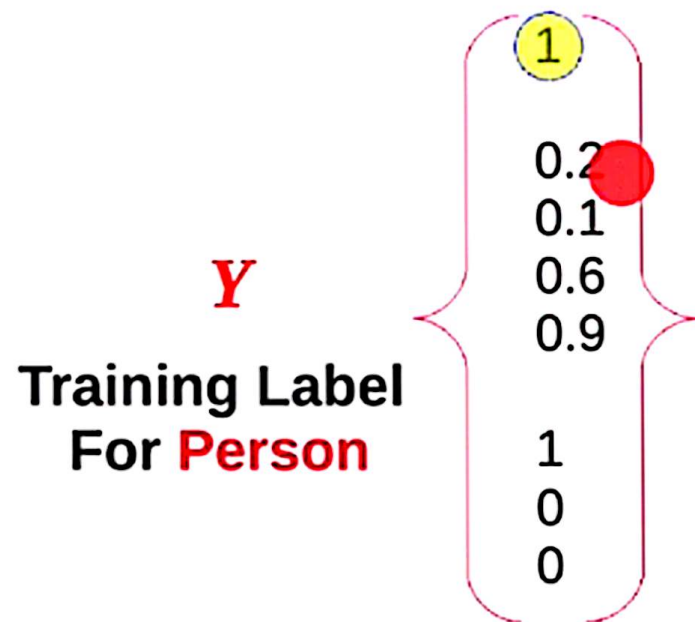
Localization and Classification

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Localization and Classification Loss Calculations

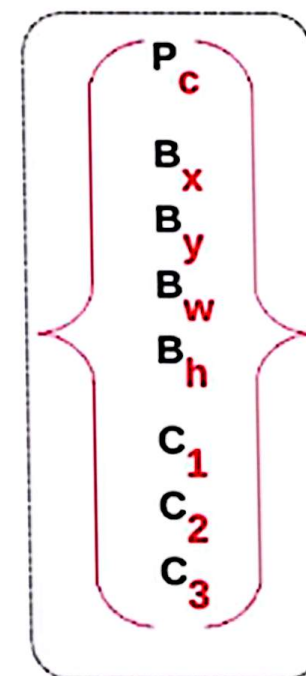
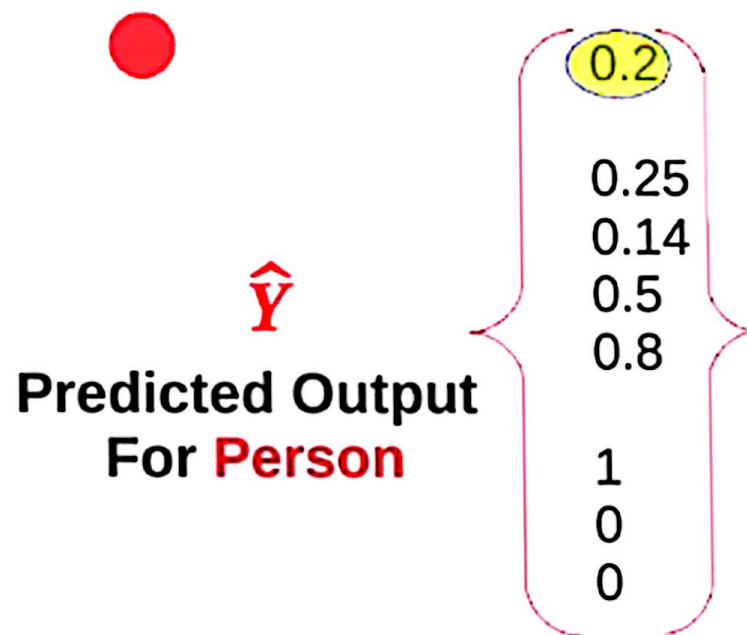
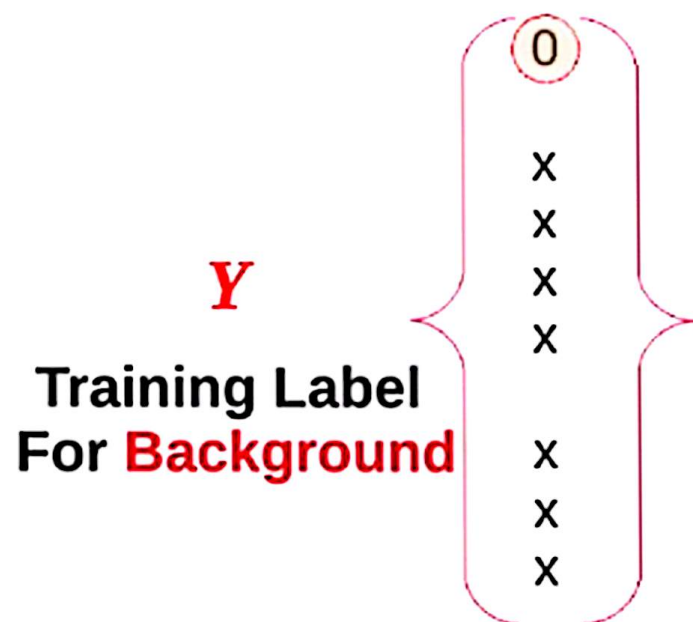
In Case of existence of Object ($Y_1 = P_c = 1$)



$$Loss = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + \dots + (y_8 - \hat{y}_8)^2 +$$

Localization and Classification Loss Calculations

In Case of existence of Object ($Y_1 = P_c = 0$)



$$Loss = (y_1 - \hat{y}_1)^2$$

Other values are **Don't Care**

Localization and Classification Loss Calculations

In Case of existence of Object ($Y_1 = P_c = 1$)

Y
Training Label
For **Person**

1

0.2
0.1
0.6
0.9
1
0
0

$\hat{Y}$
Predicted Output
For **Person**

1

0.25
0.14
0.5
0.8
1
0
0

P_c

B_x

B_y

B_w

B_h

C_1

C_2

C_3

Squared Error Loss Function

$$Loss = (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + (y_3 - \hat{y}_3)^2 + \dots + (y_8 - \hat{y}_8)^2 +$$

Other Loss Functions that May be used: **Log Error Loss** Function (for softmax Output C_1, C_2, C_3)
Logistic Regression Loss (for P_c)

Object Detection

(Detection of **Multiple** Objects)

Detection of Multiple Objects

[1] Sliding Window

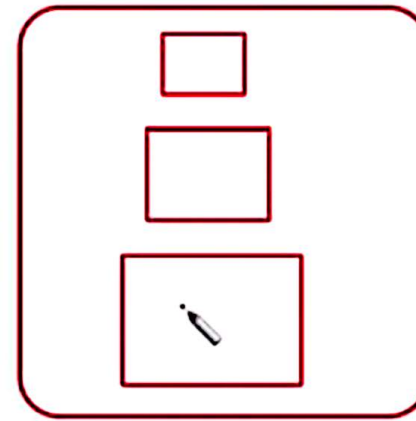
[2] You Only Look Once (YOLO)

[3] Regional CNN (RCNN)
(RCNN – Fast RCNN – Faster RCNN)

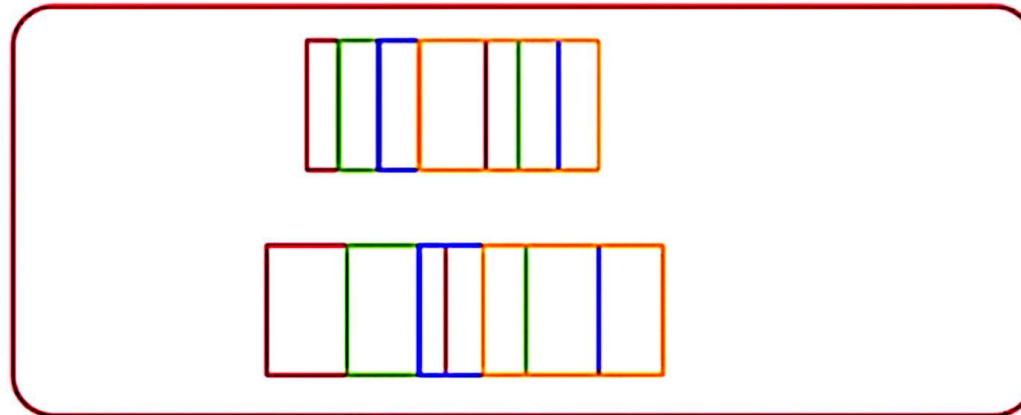
[1] Sliding Window Technique

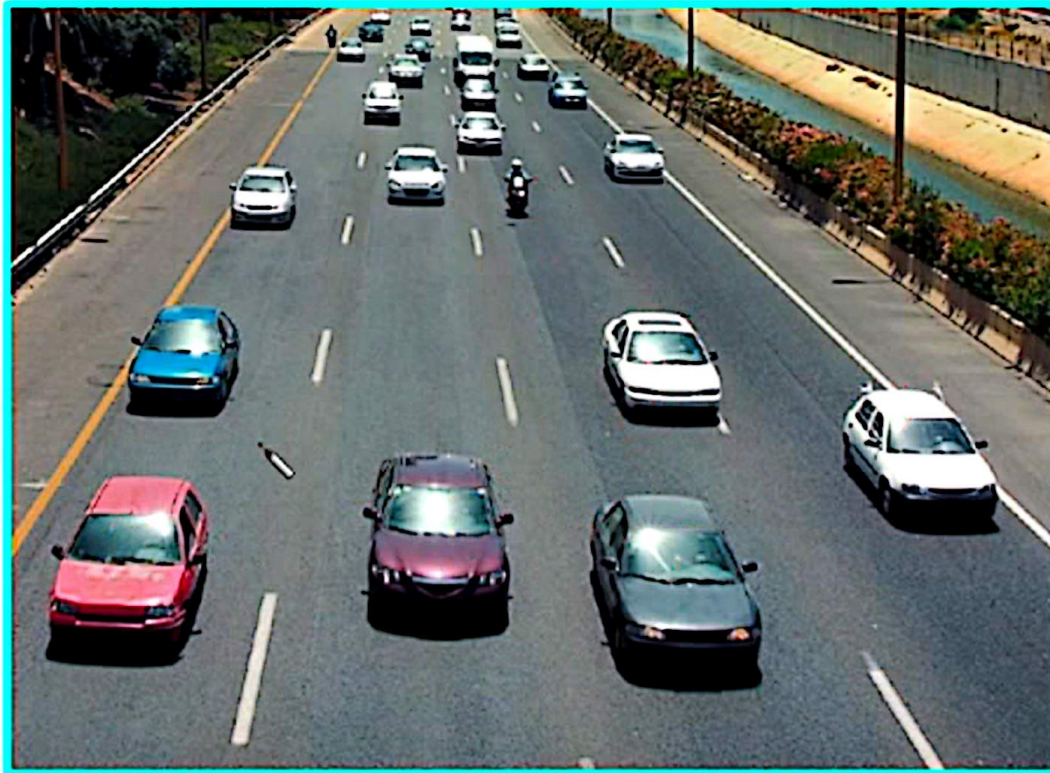
Sliding Window Technique

Use Different Window Sizes



Select the required Stride (precision)



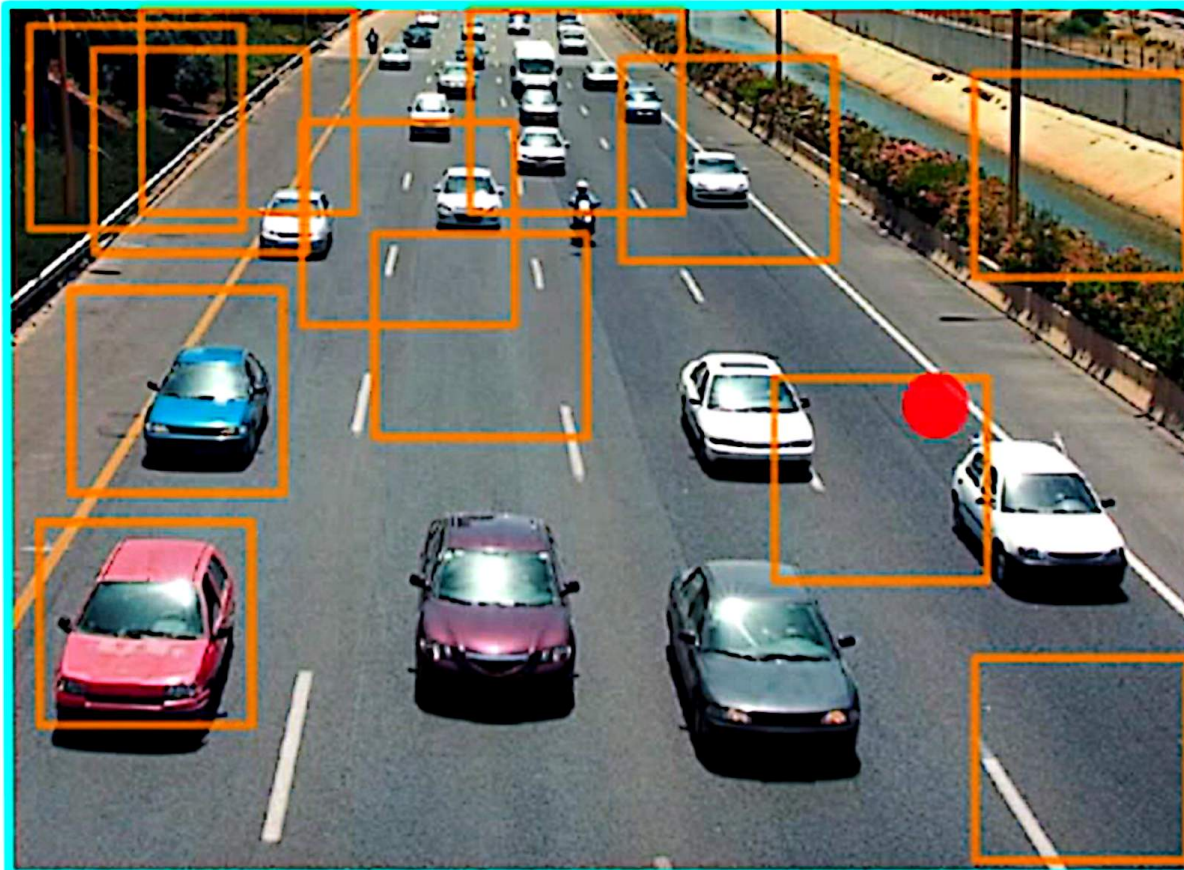


Input Image

Scan using **MULTIPLE** Windows sizes

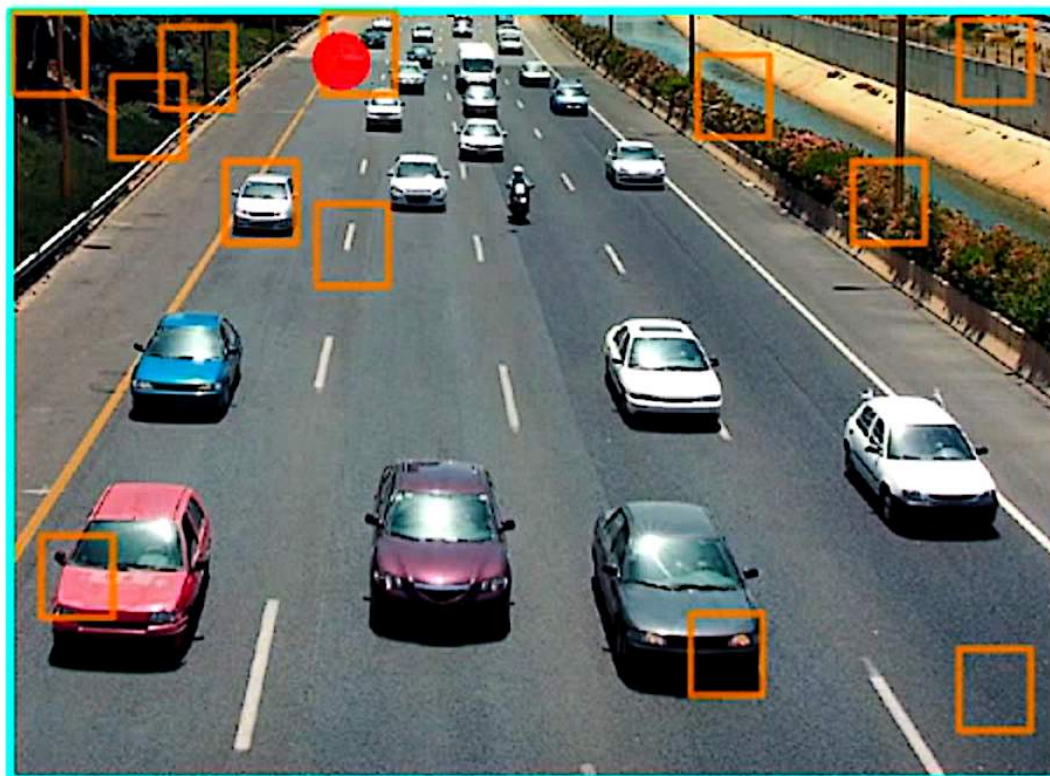


Window
of Size 1



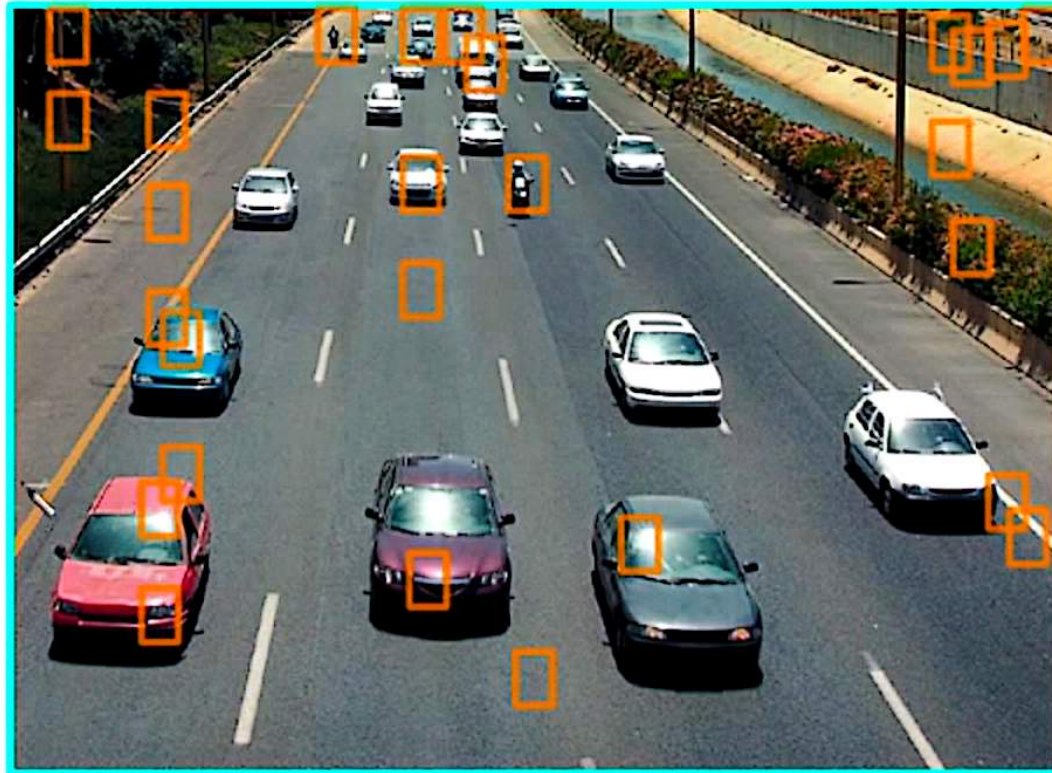


Window
of Size 2

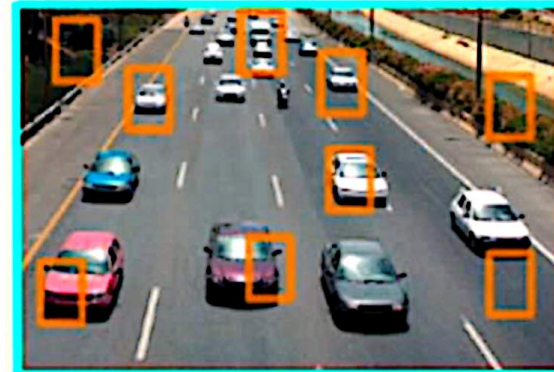
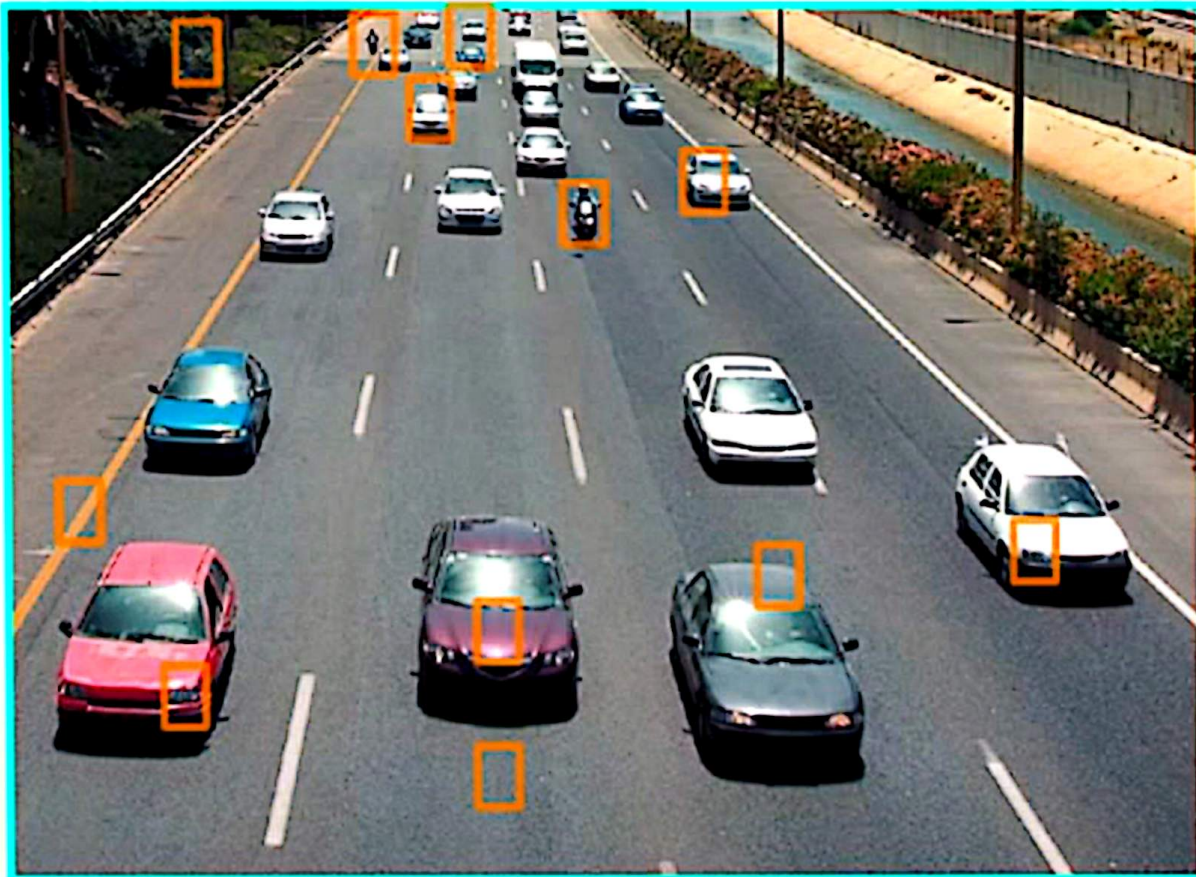




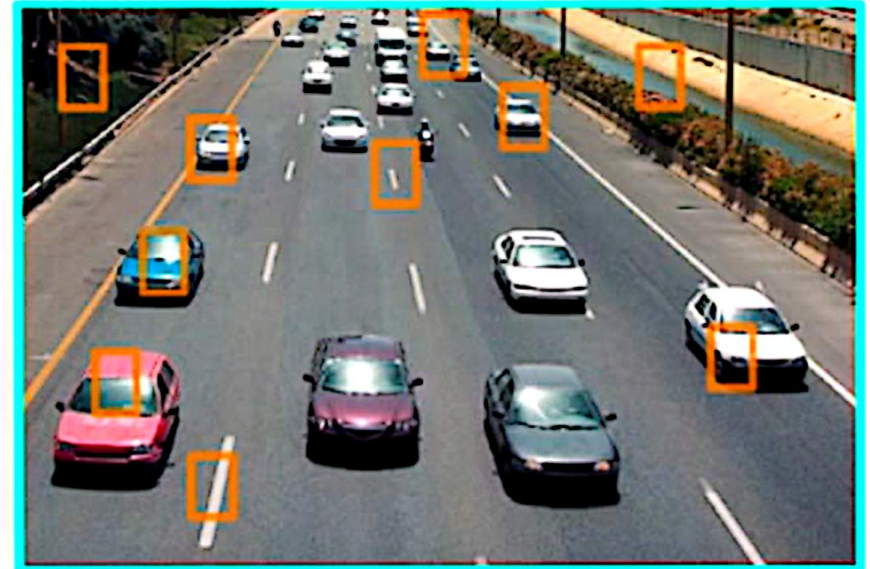
Window
of Size 3

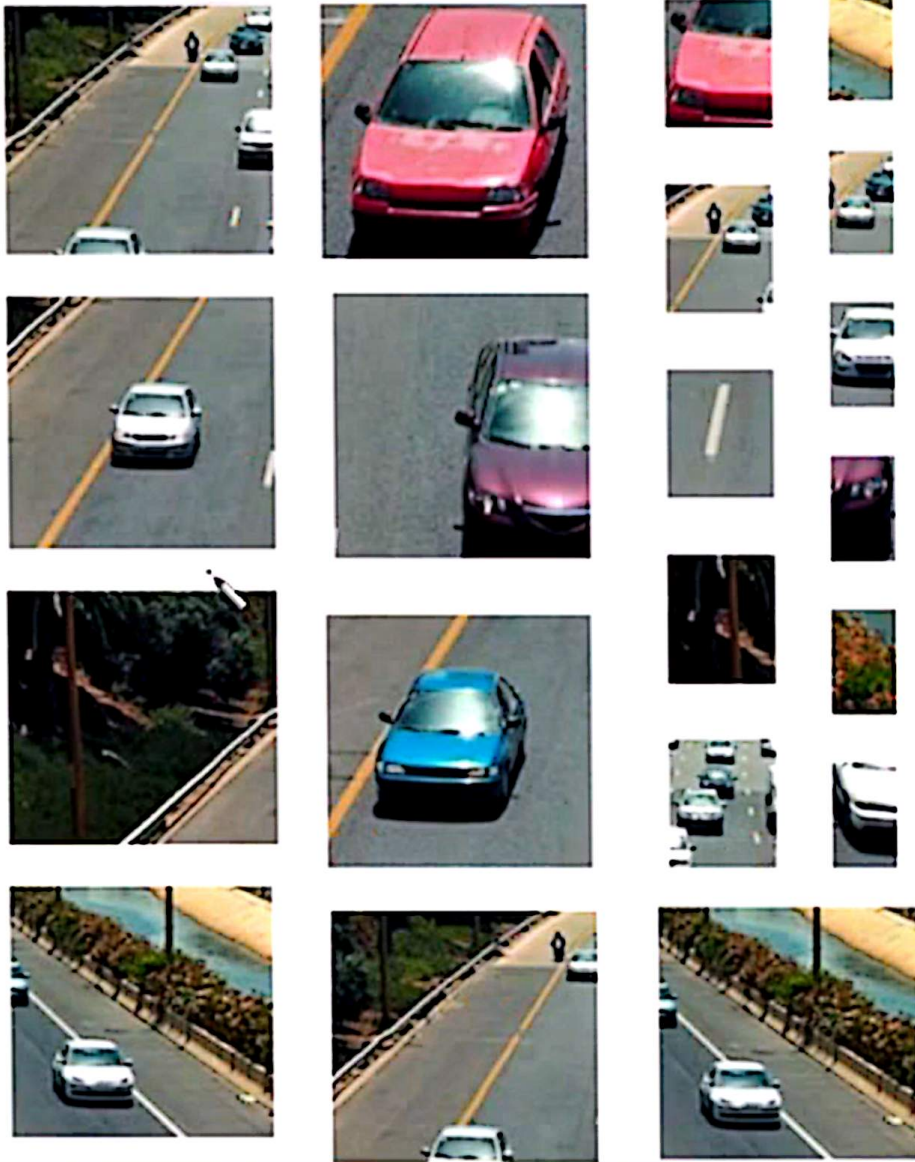


Alternatively, One Window size, Multiple Image Scales



Window
Size



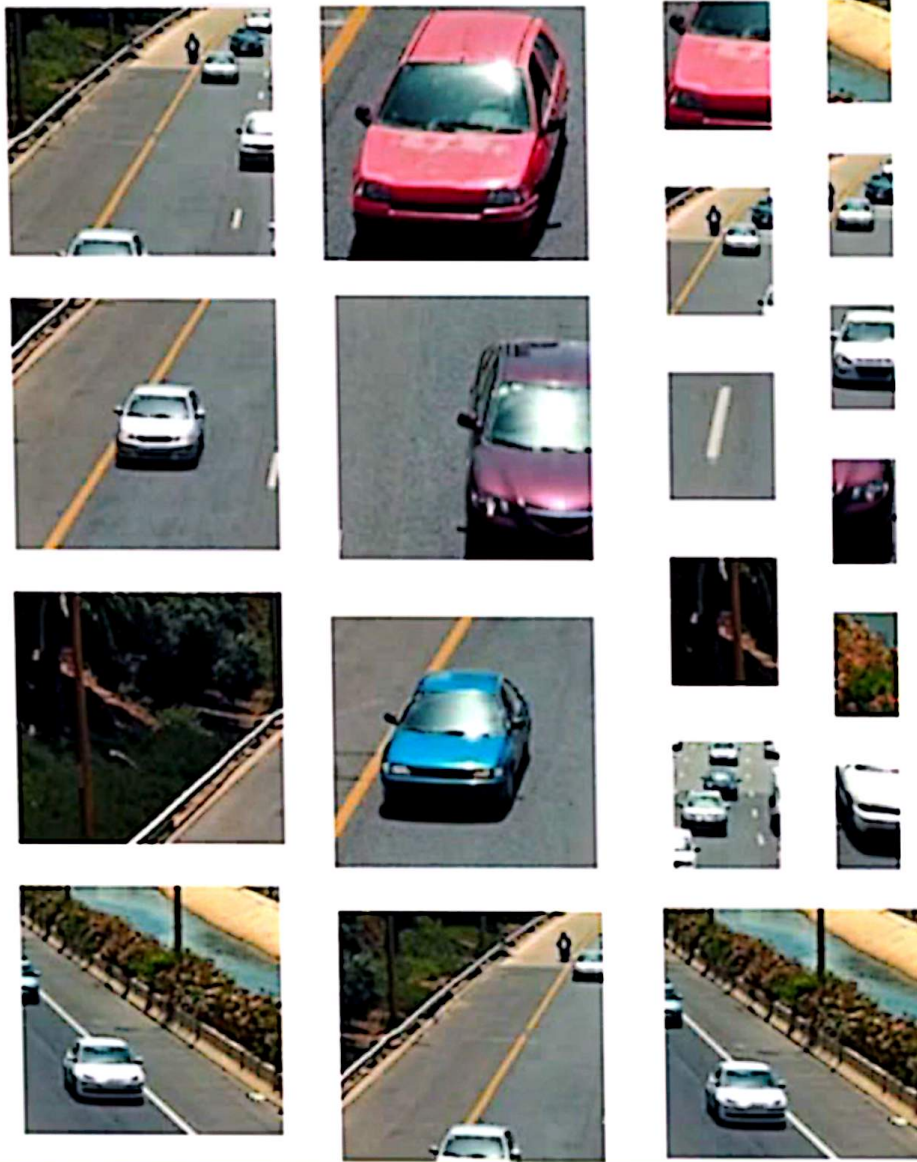


**Object
Classification**



$\begin{pmatrix} \dots \\ \dots \\ \dots \\ \dots \\ \dots \\ \dots \\ \dots \\ \dots \\ \dots \\ \dots \end{pmatrix}$

***Objects
detected as cars***



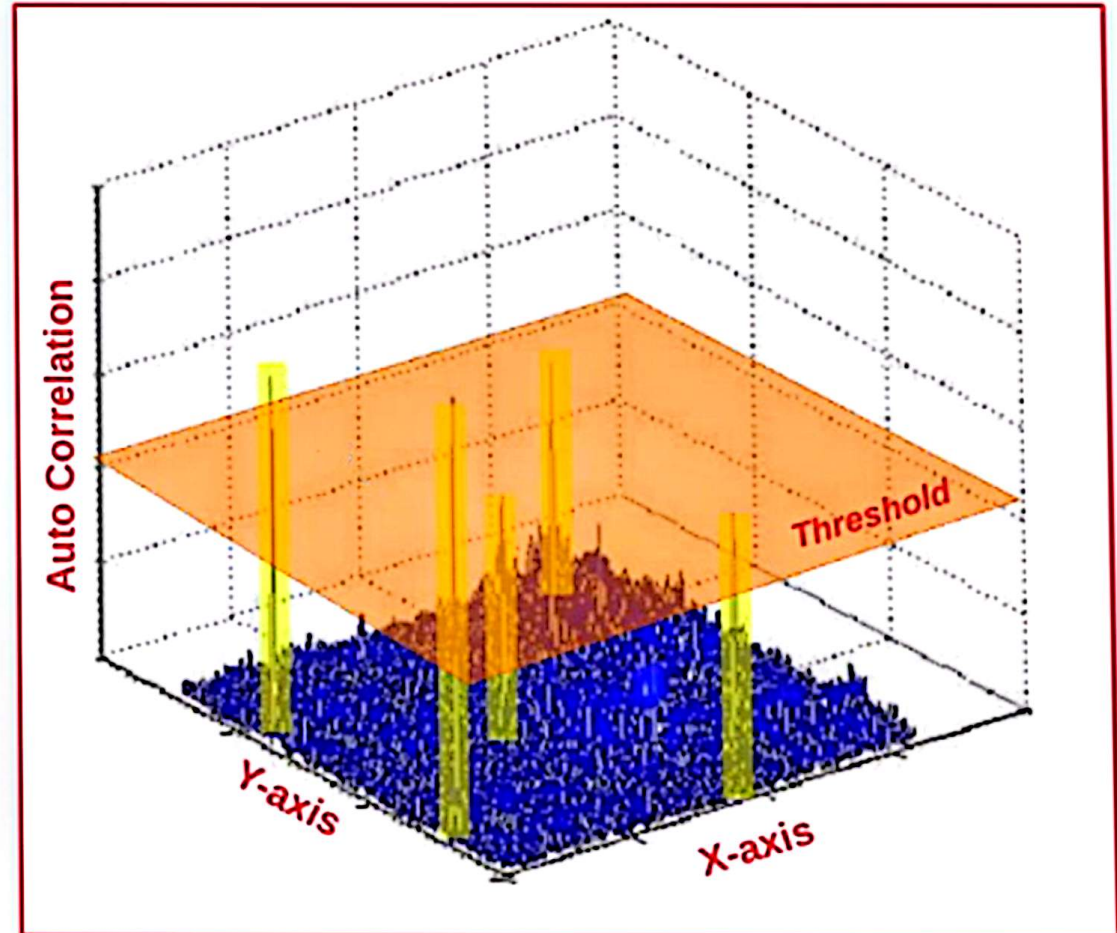
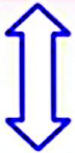
Object Classification and Localization

Classical Techniques

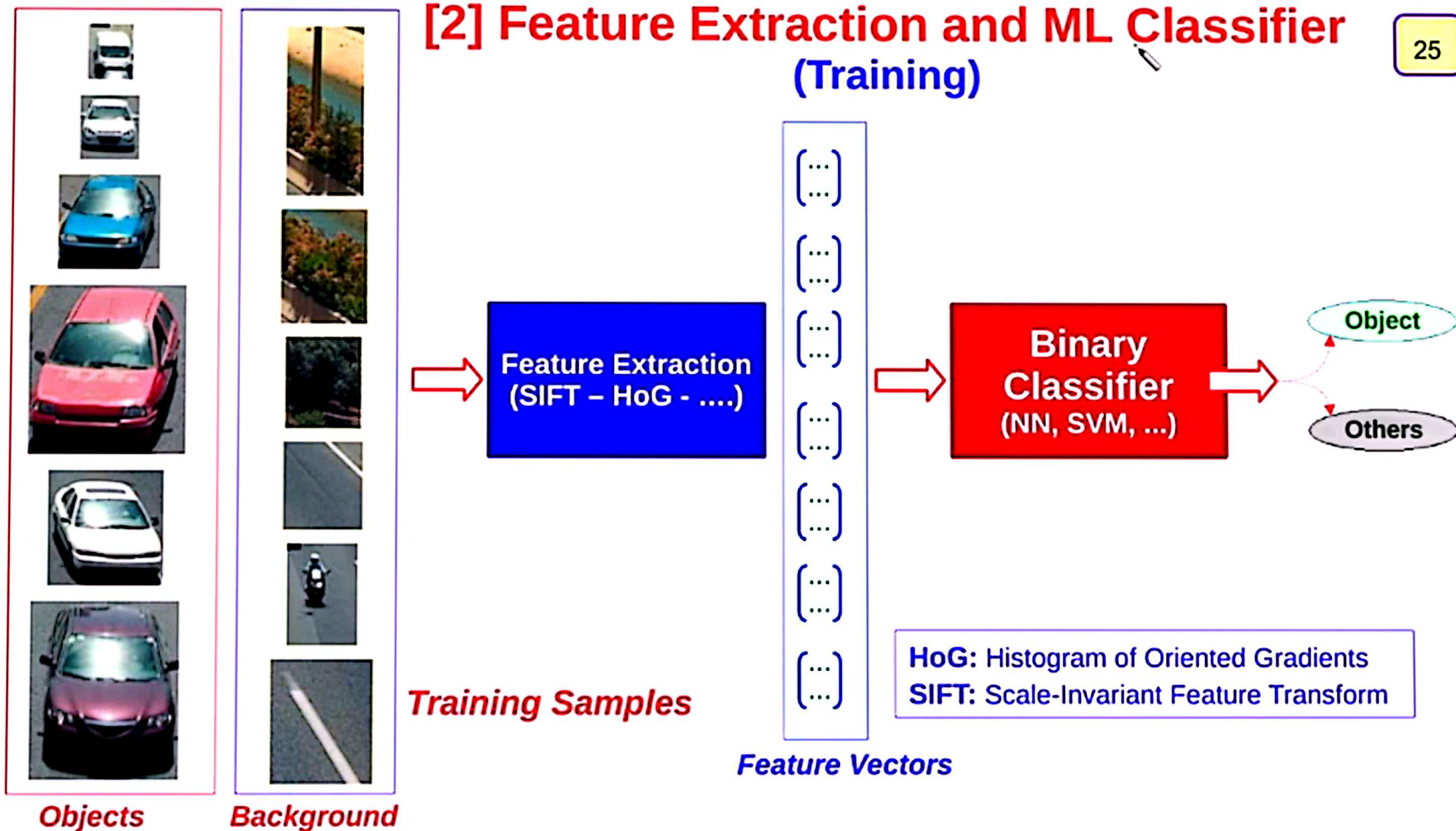
Deep Learning Techniques

Objects detected as cars

[1] Template Matching using Auto Correlation



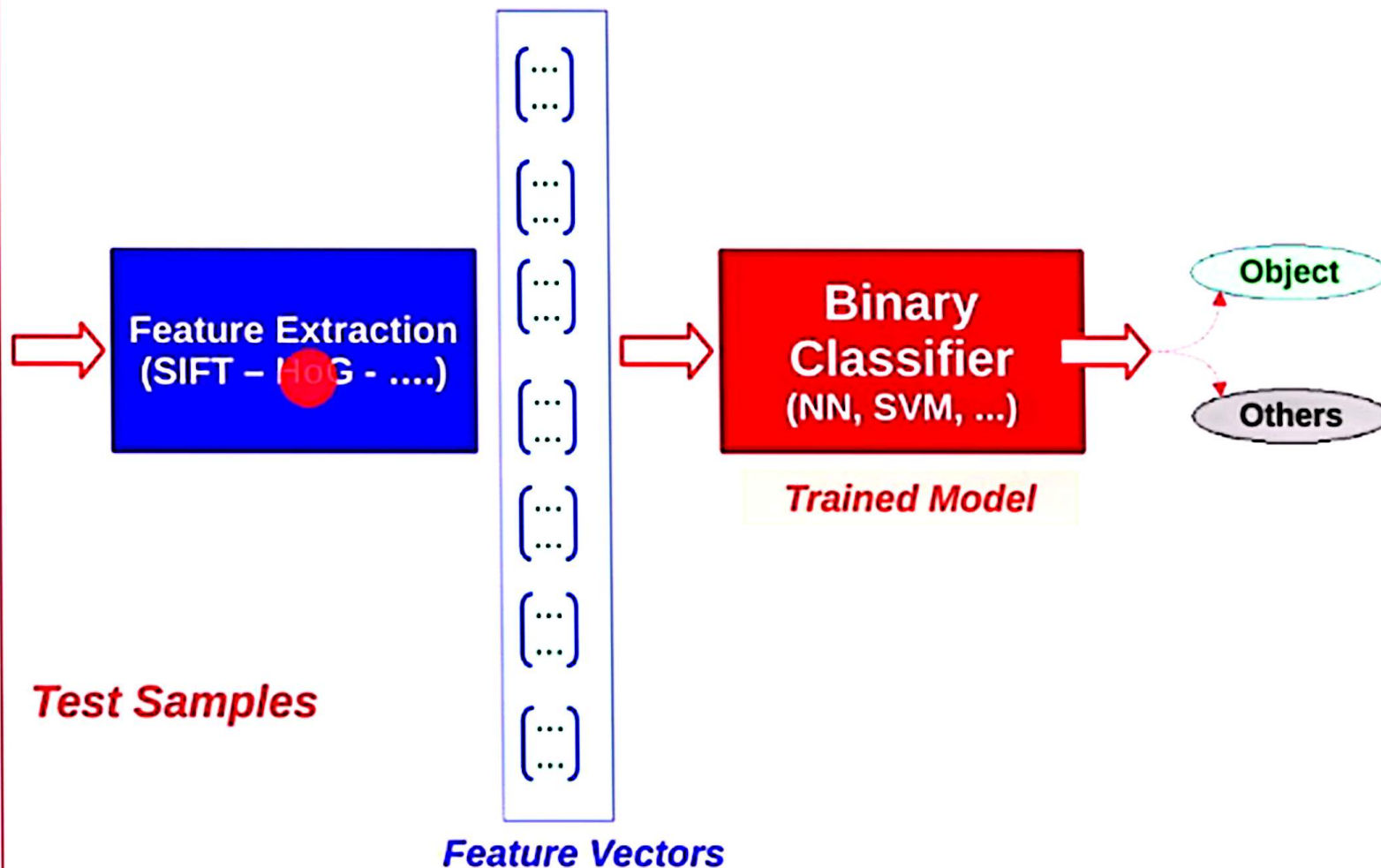
[2] Feature Extraction and ML Classifier (Training)



[2] Feature Extraction and ML Classifier (Inference)

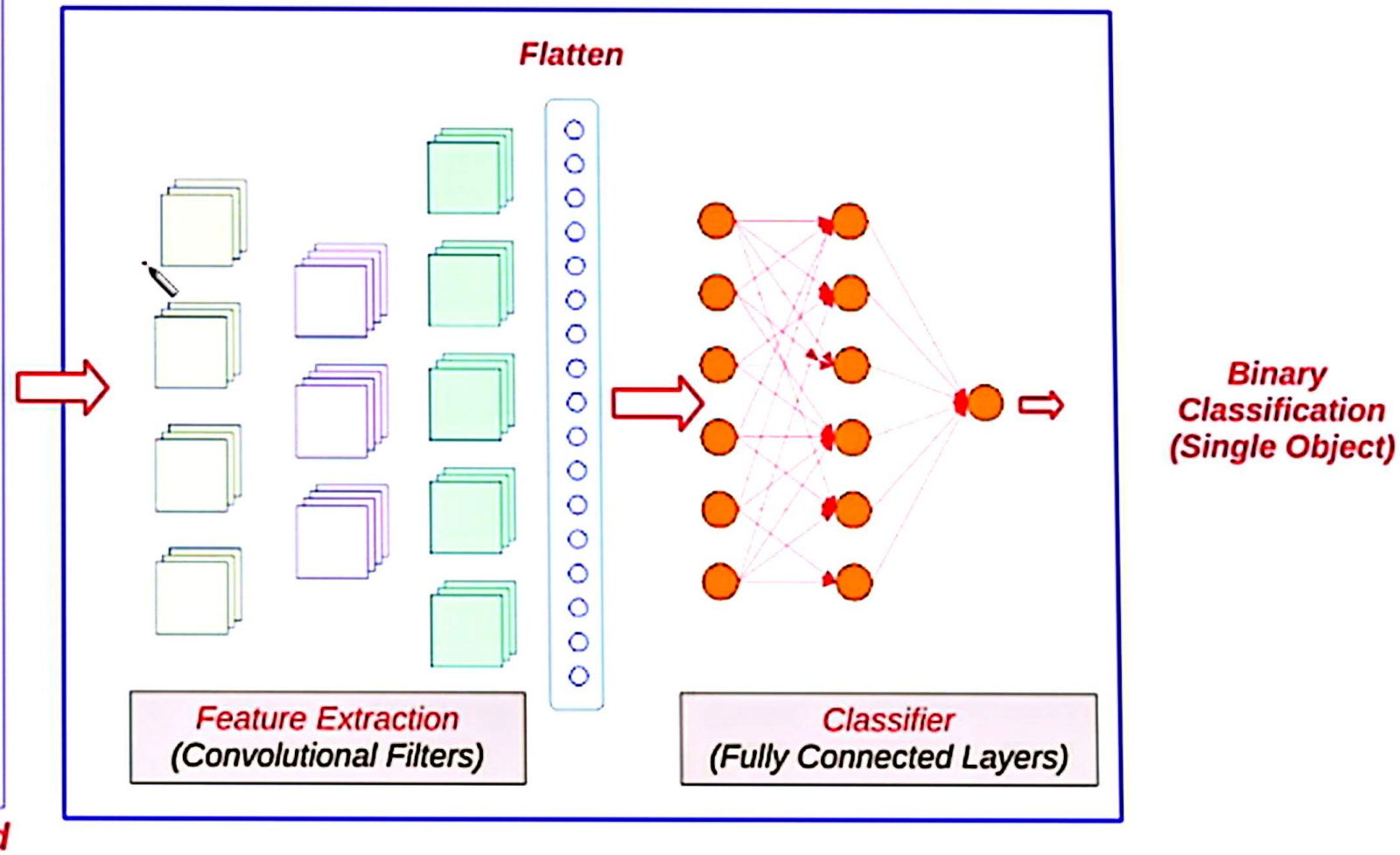


Images from Sliding Window



[3] Convolutional Neural Network (Training - Single Object)

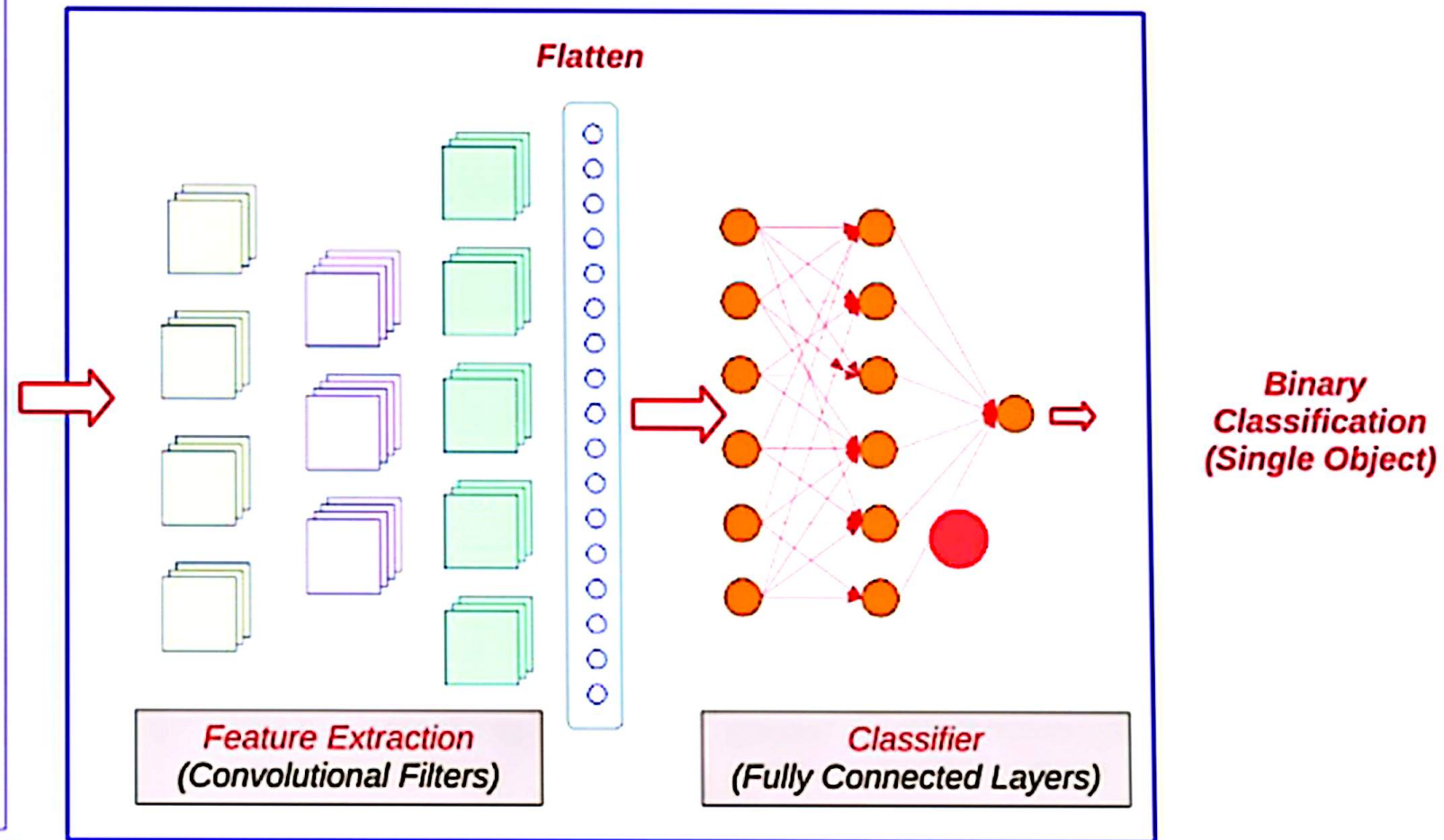
CNN



**Binary
Classification
(Single Object)**

[3] Convolutional Neural Network (Training - Single Object)

CNN

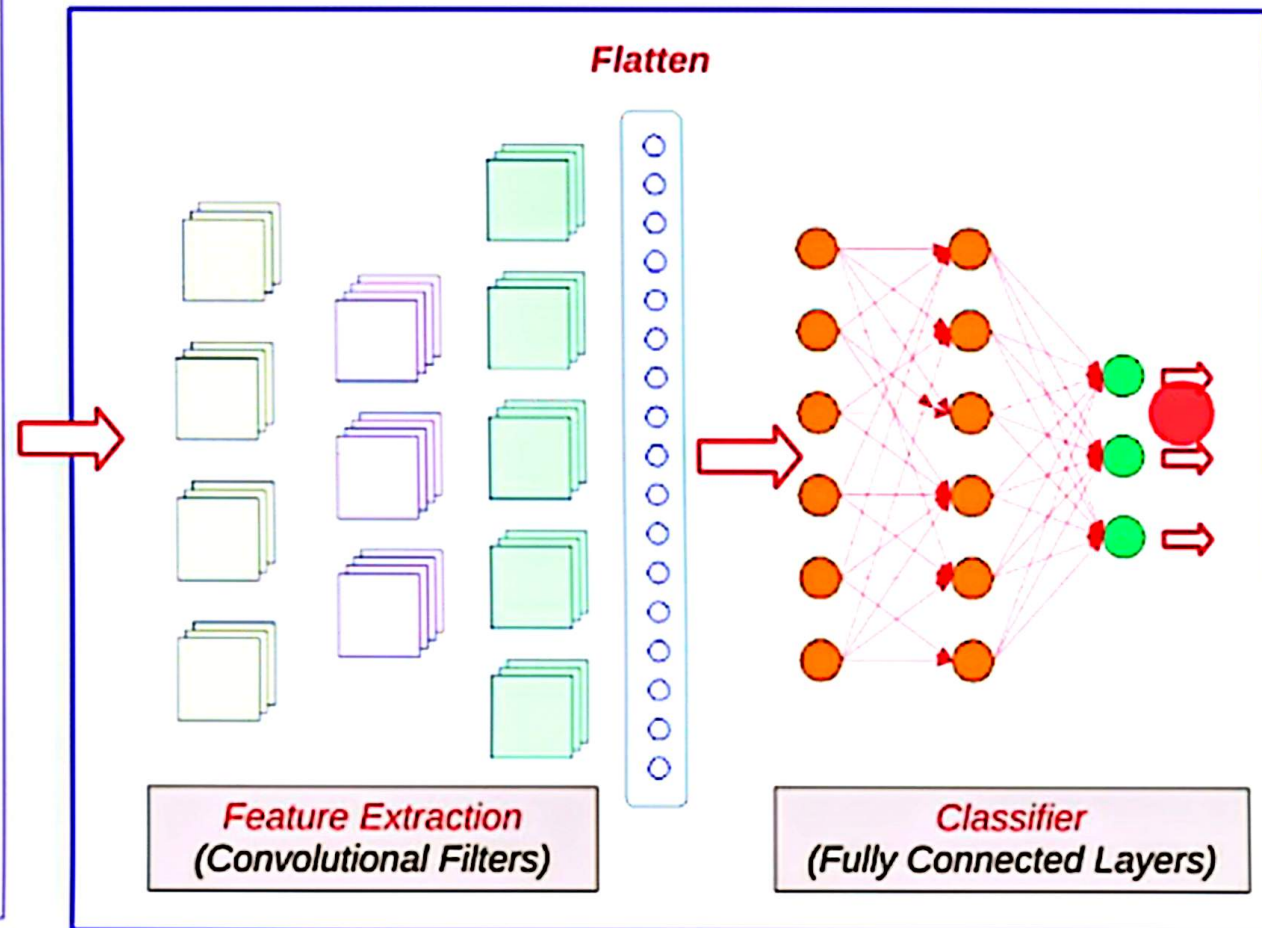


Objects

Background

[3] Convolutional Neural Network (Training - Multiple Objects)

CNN



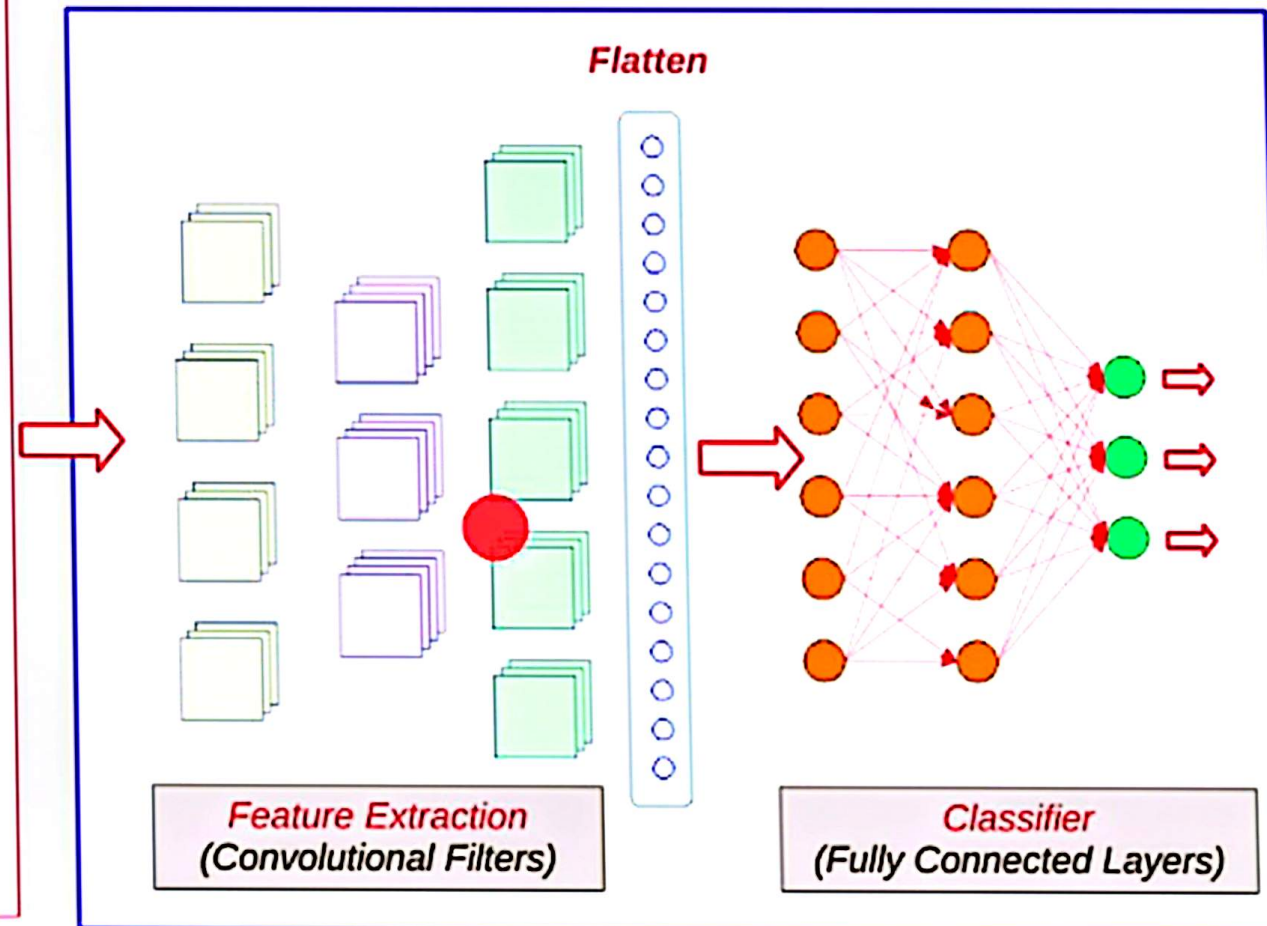
Objects

Background

[3] Convolutional Neural Network (Inference - Multiple Objects)

Test Samples

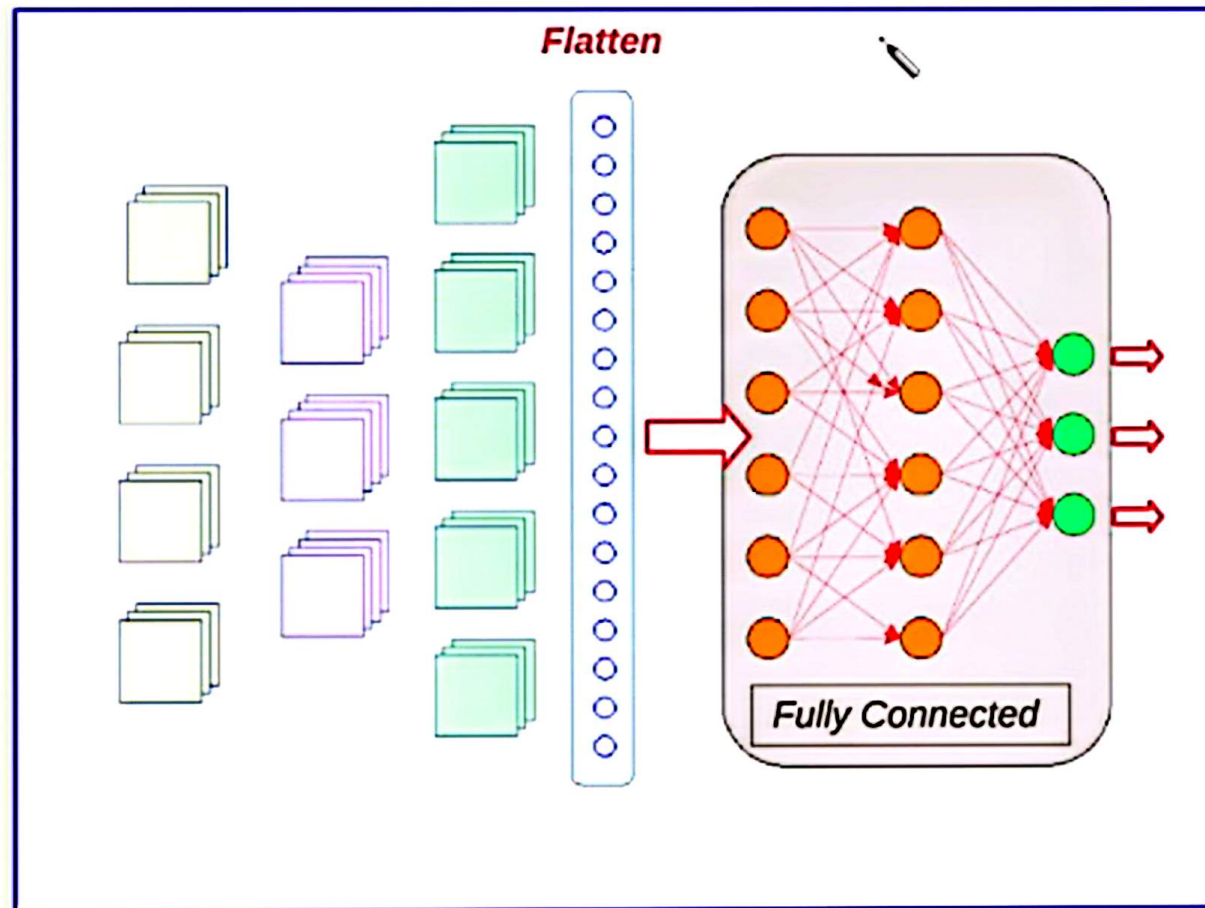
CNN

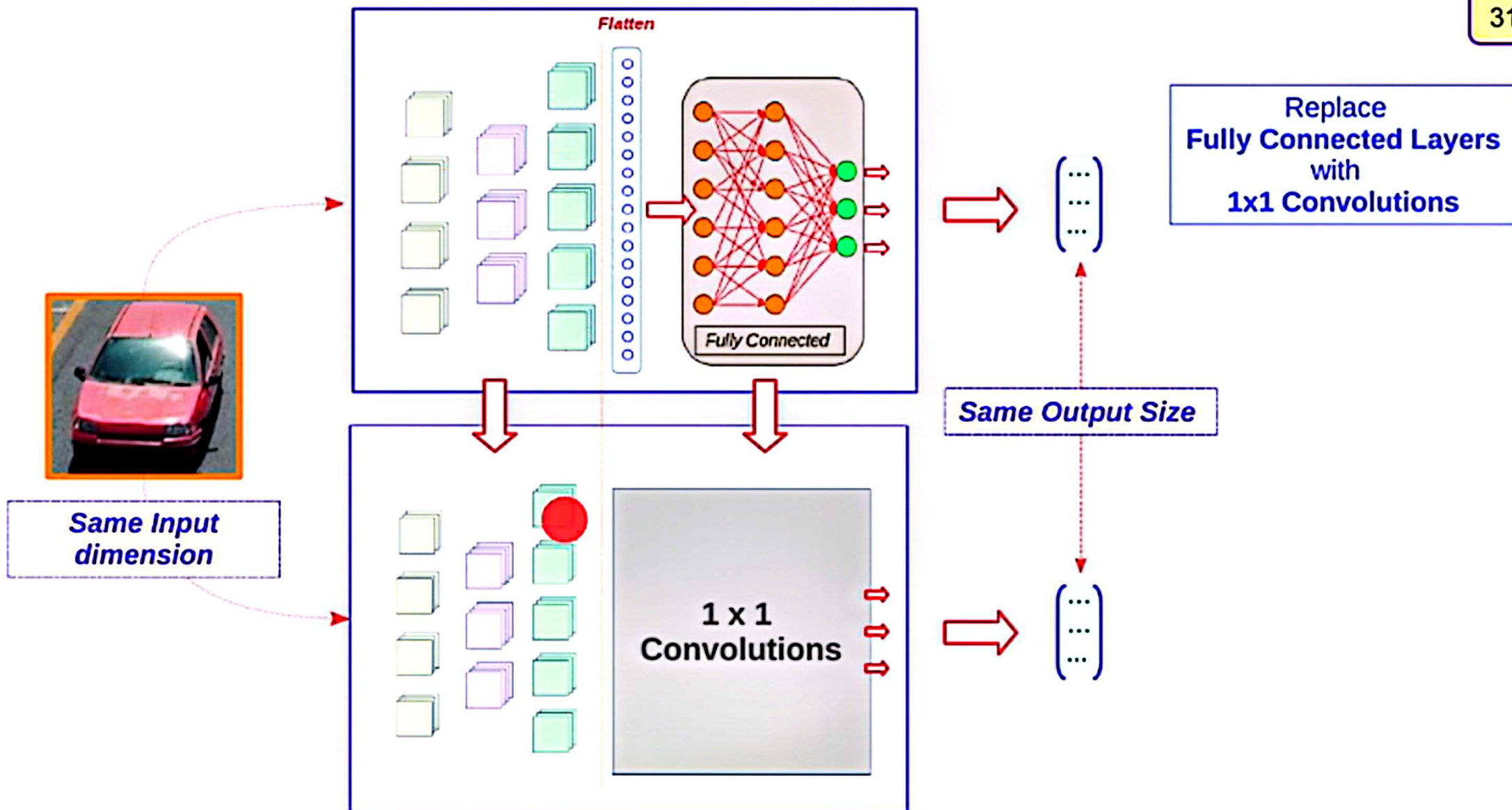


Images from Sliding Window

[3] Implementation of Fully Connected Layers with Convolutional Layers

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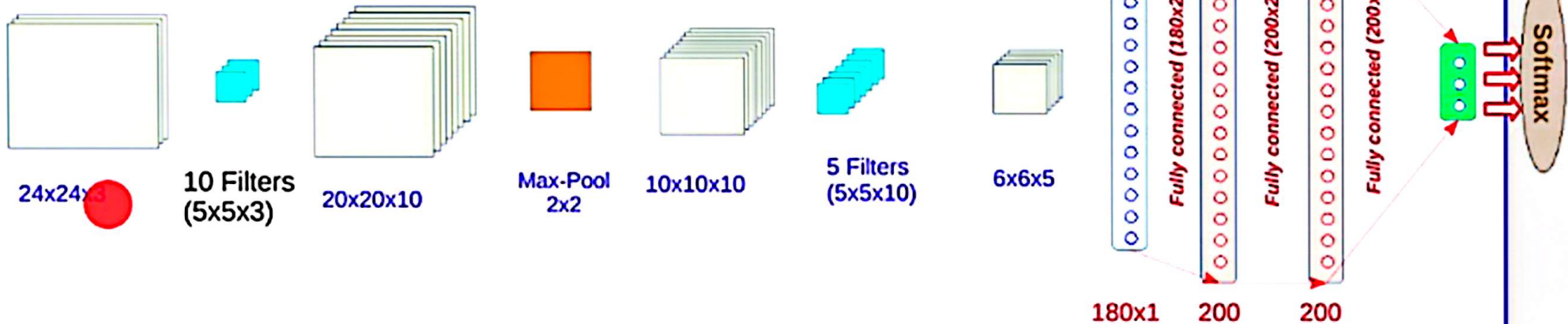




[3] Example: Implementation of FC Layers with Conv. Layers

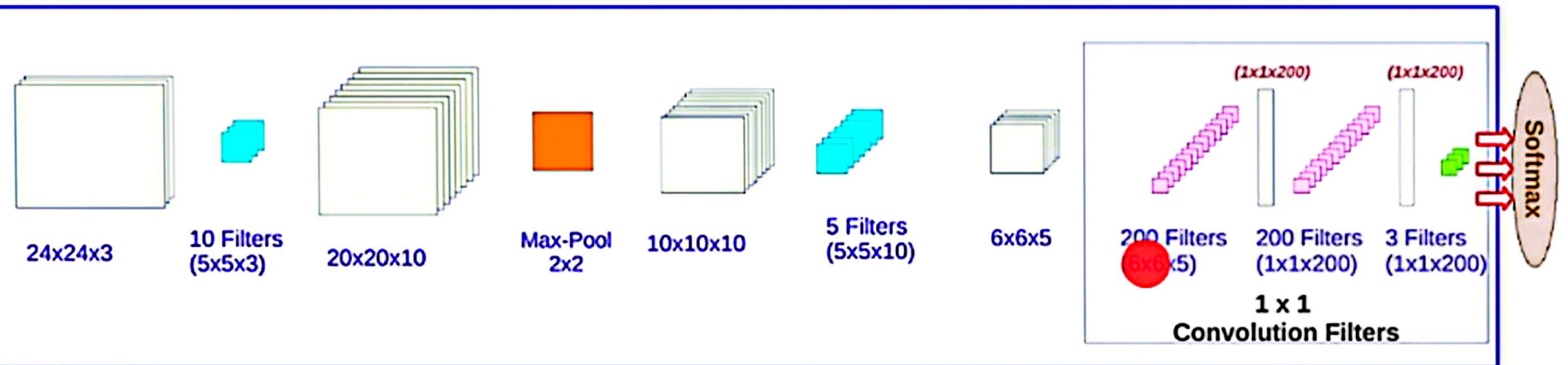
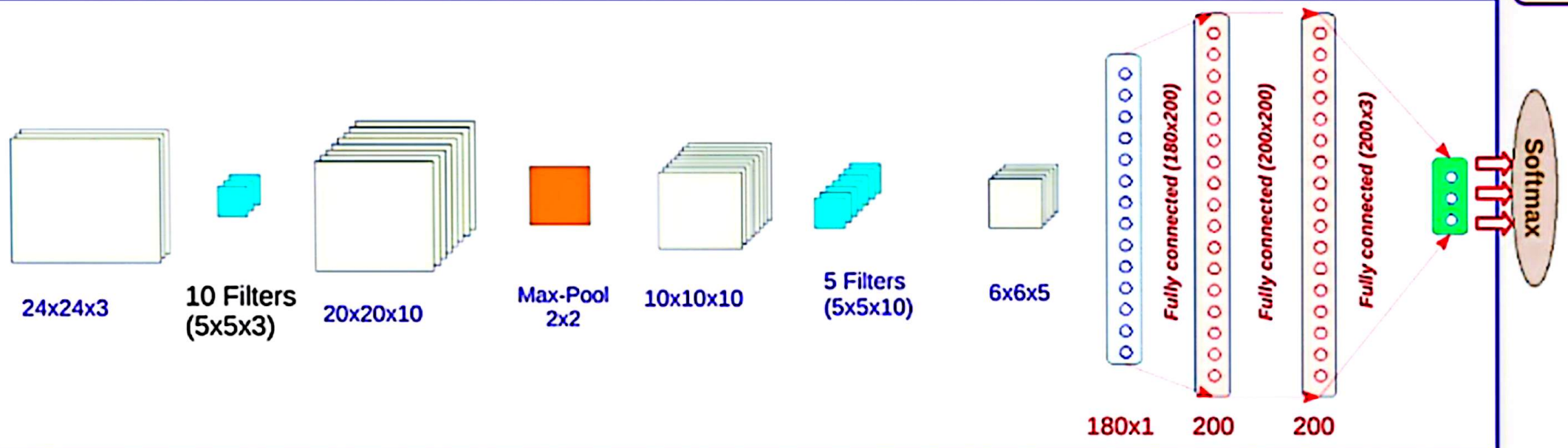
32

Stride=1, No Padding



[3] Example: Implementation of FC Layers with Conv. Layers

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Apply CNN for ALL Sliding windows images



100x100

Let:
Sliding Window Size = 21x21

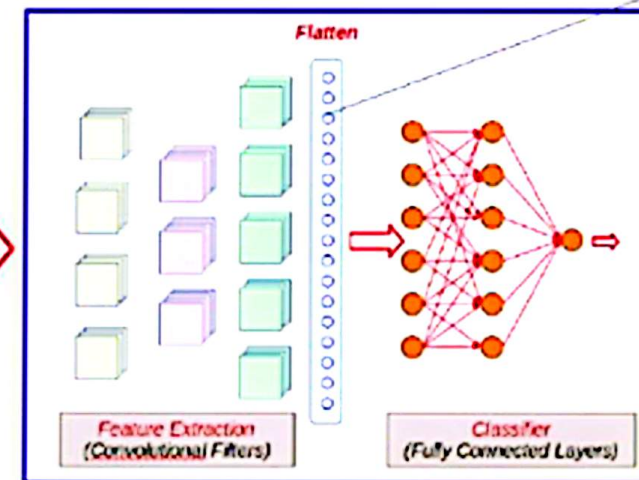
21 x 21

Scan the 100x100 image
using 21x21 sliding window
With Stride=1

Results in 80x80=1600 images

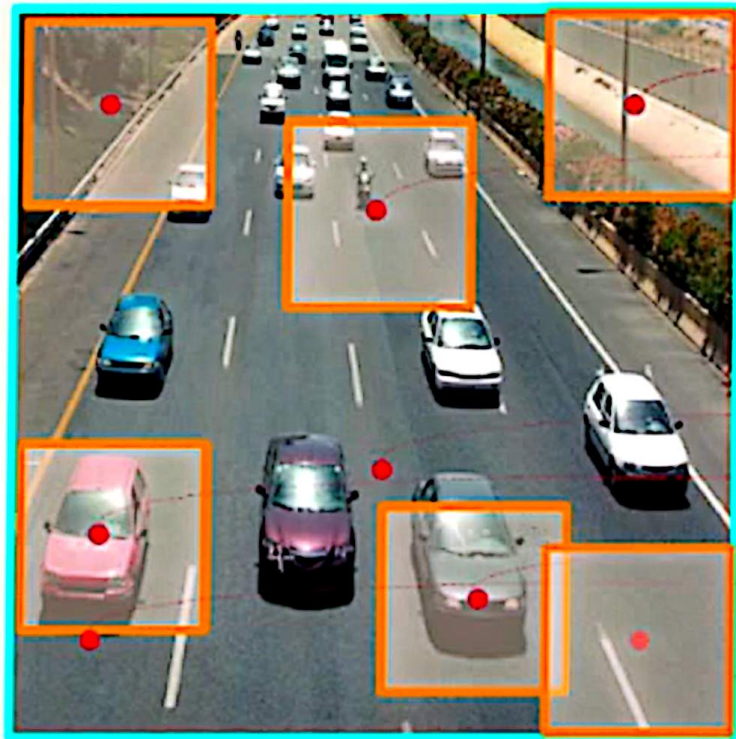


21x21



Classify 1600 images (one by one)

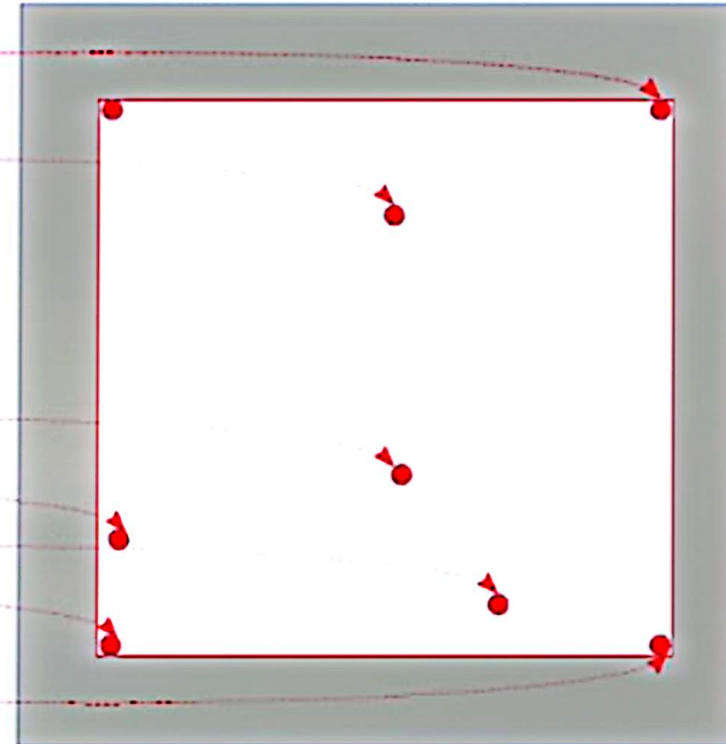
Apply CNN for ALL Sliding windows images



100x100

Apply Filter of size **21x21**
with *stride= 1, no padding*

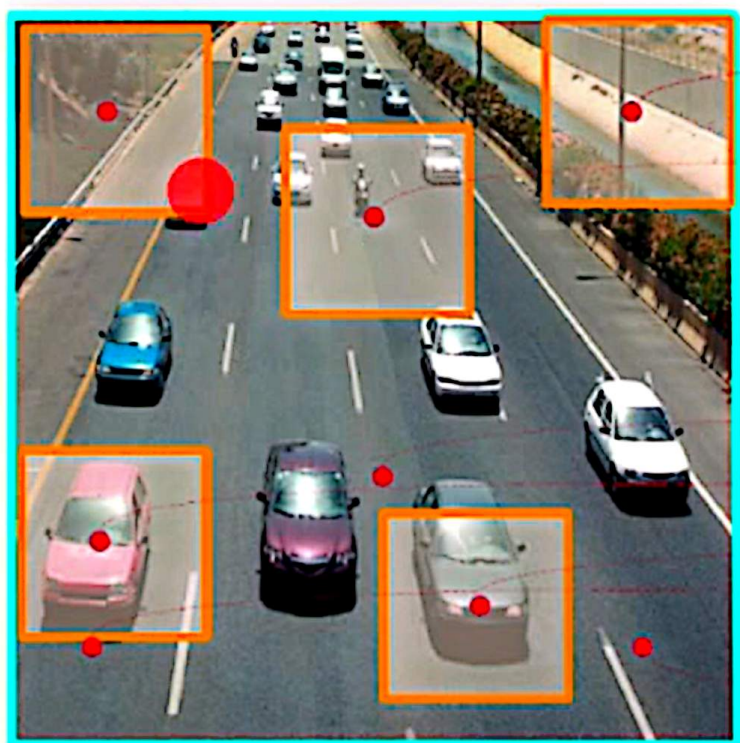
21x21



80 x 80

Each output represents value of applying
the filter on the underneath 21x21 image

Apply CNN for ALL Sliding windows images



100x100

Apply Filter of size 21x21
with stride 1, no padding

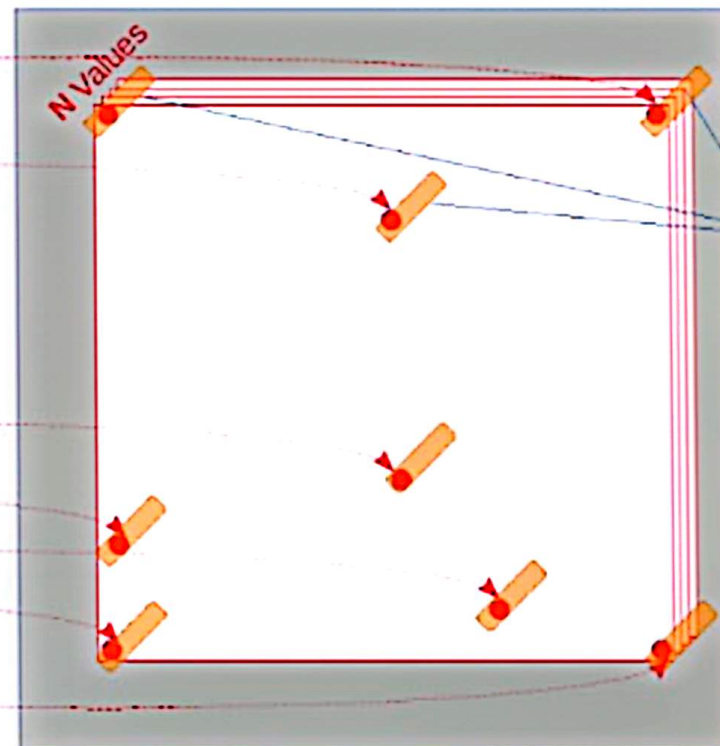
21x21

21x21



21x21

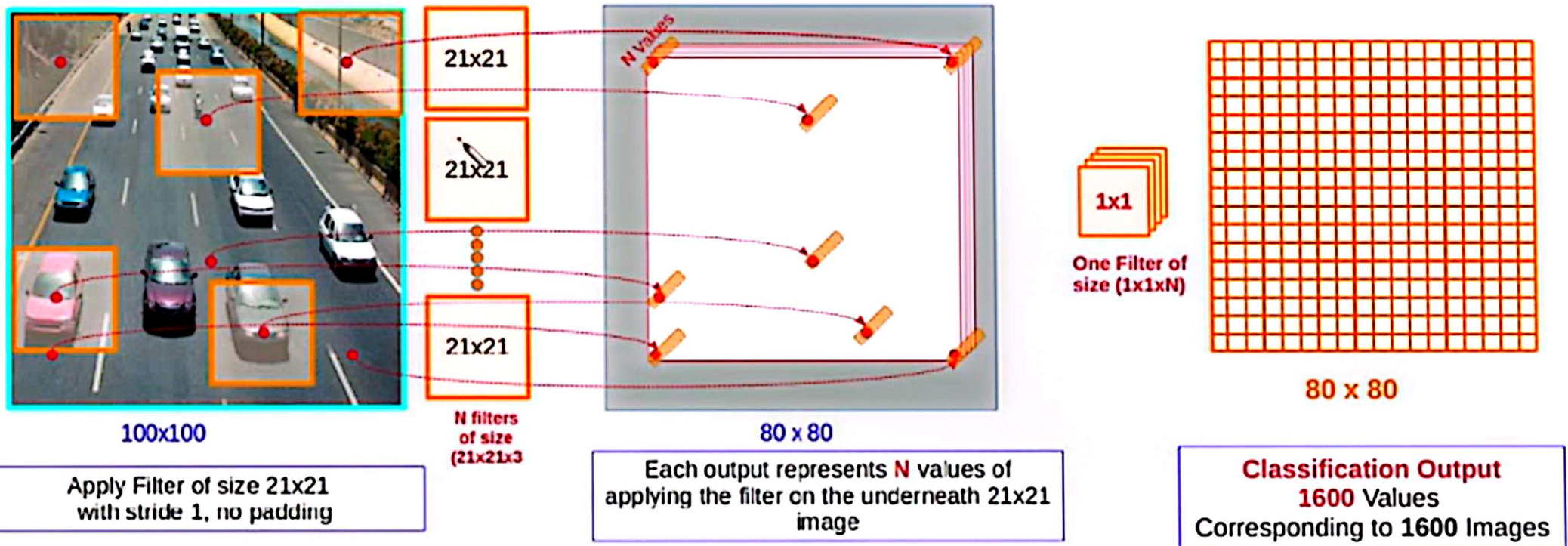
N filters
of size
(21x21x3)



80 x 80

Each output represents **N** values of
applying the filter on the underneath 21x21
image

Apply CNN for ALL Sliding windows images

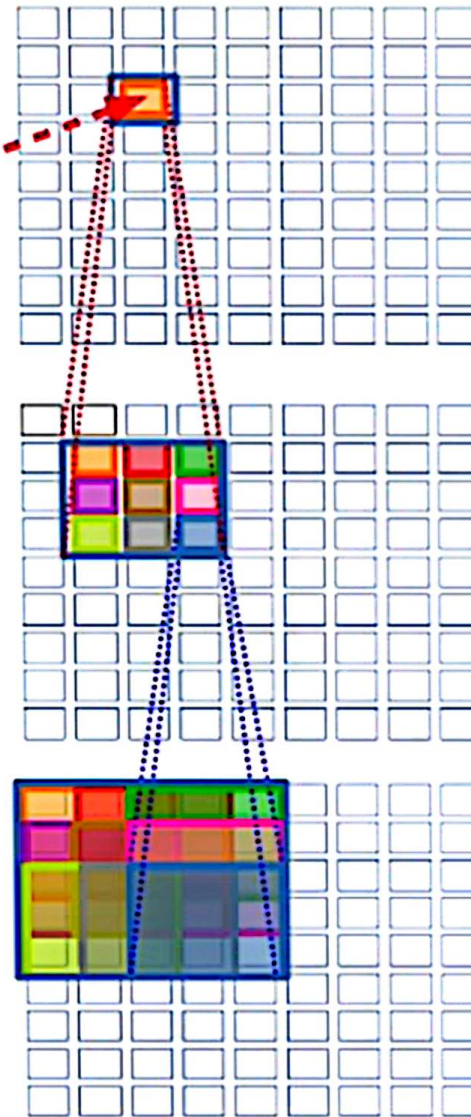
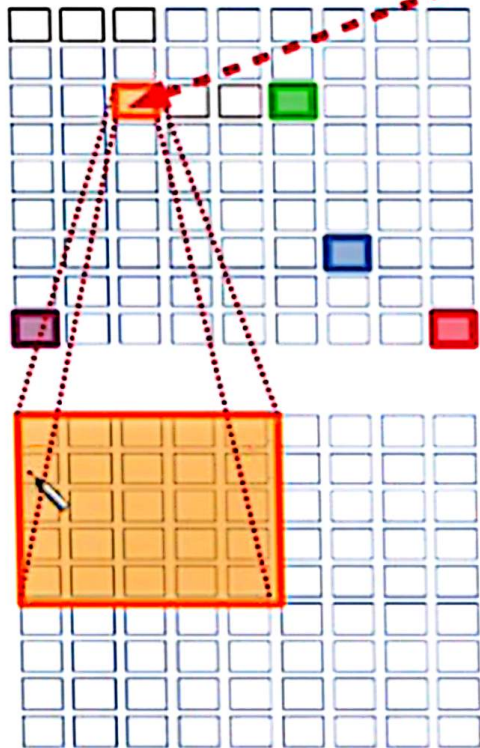


Use shared Calculations to parallelize classification of 1600 images (one shot)

Apply 21x21xN filter is time consuming !!!

Review: Filters Factorization

5 x 5 Convolution
Using two Layers 3 x 3 Convolution

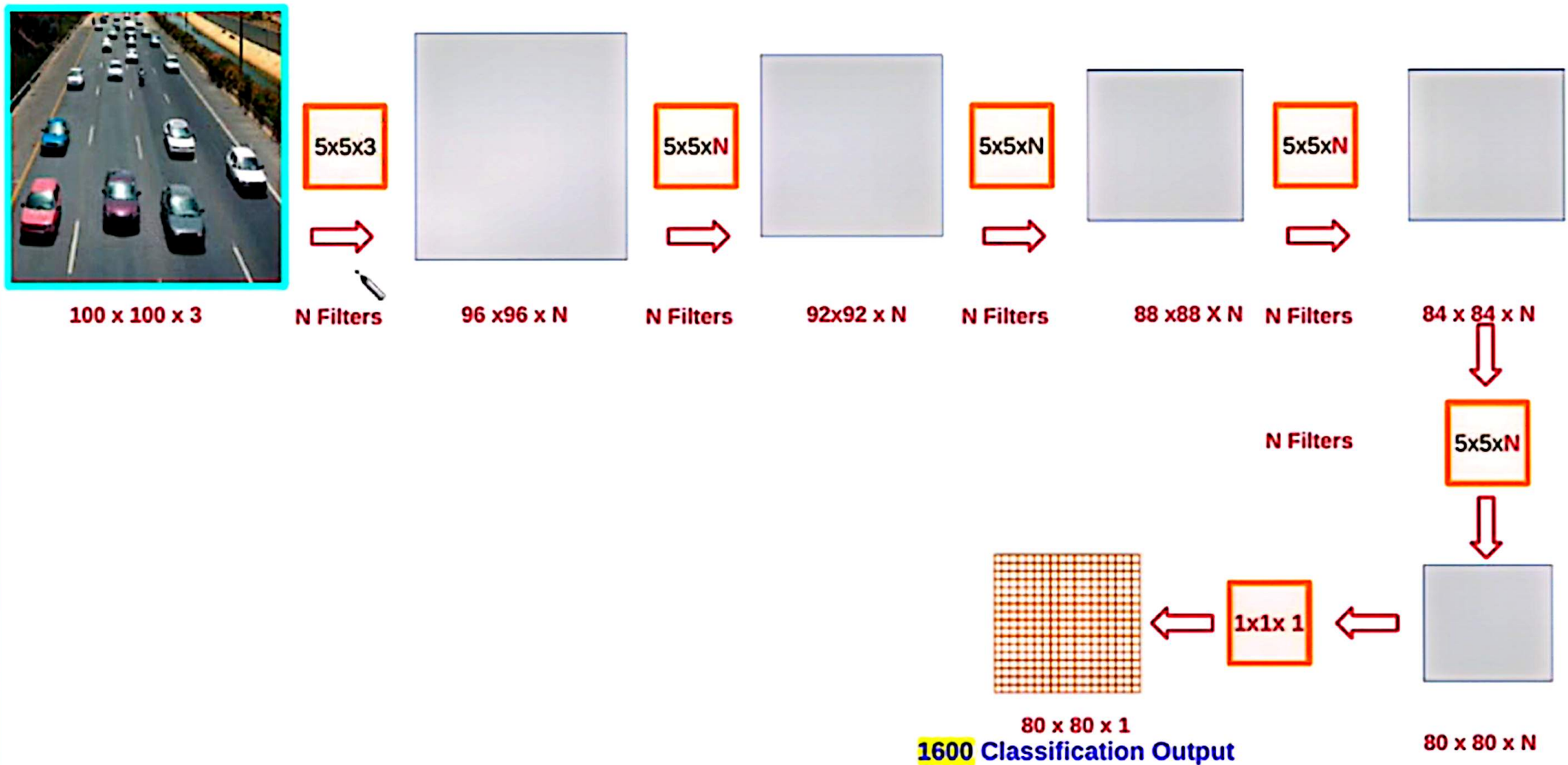


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Output Representing
5x5 sub-image

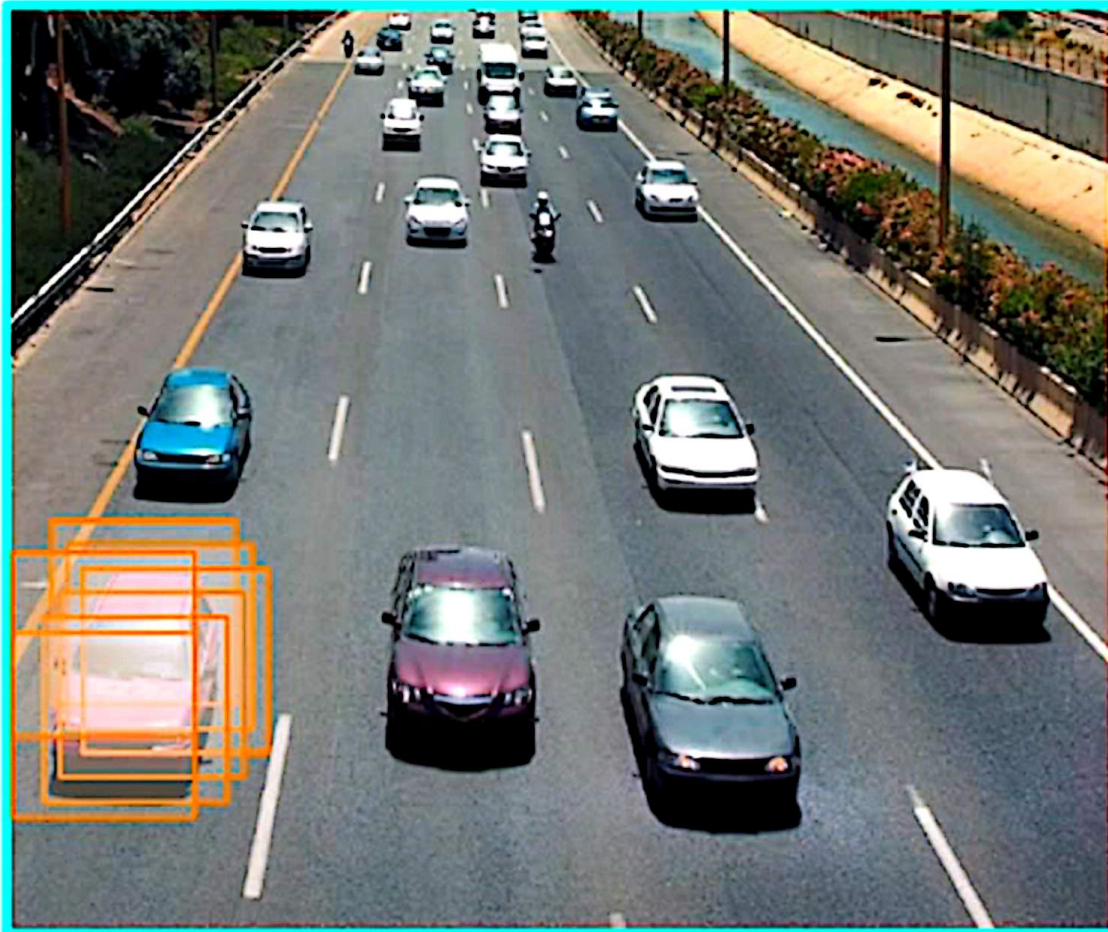
Calculations are
reduced to 72%

Factorize the 21x21 filter using 5x5 filters



Problem of OVERLAPPED Detections

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**Same object
can be detected with
Multiple windows !!!**

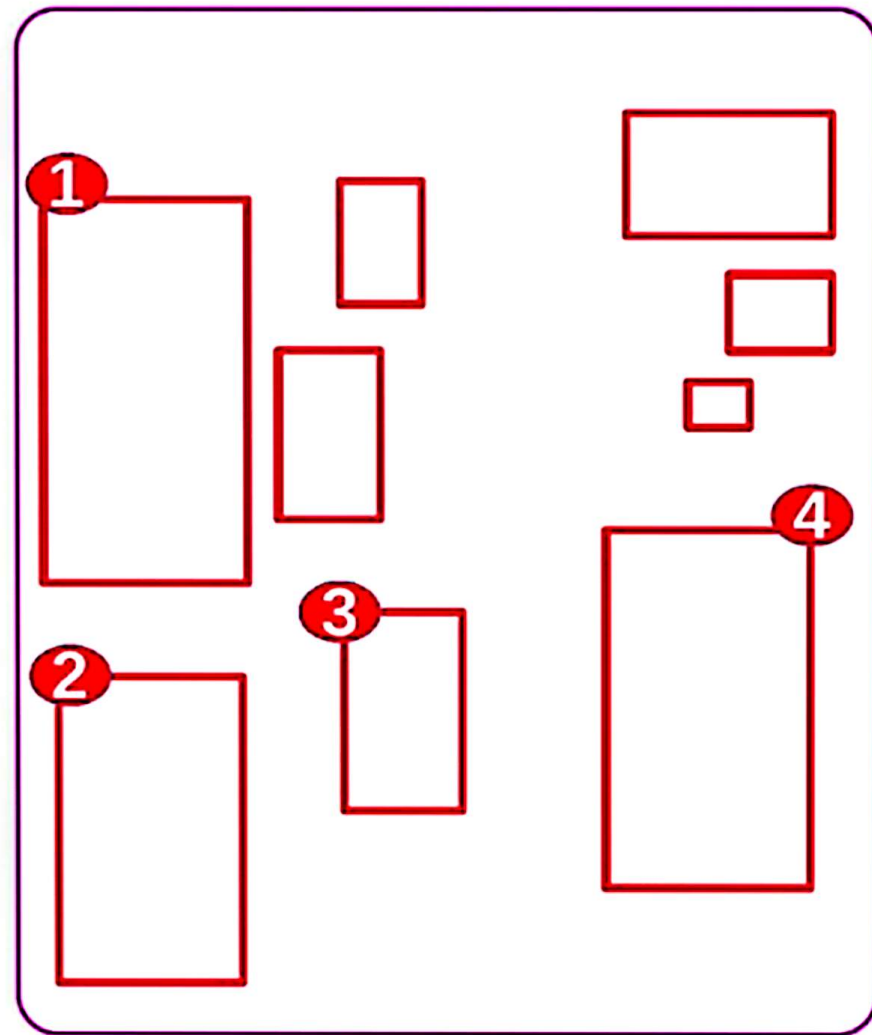
Use
**Non-Maxima Suppression
Algorithm**
To resolve overlapping problem

Sliding Window Technique

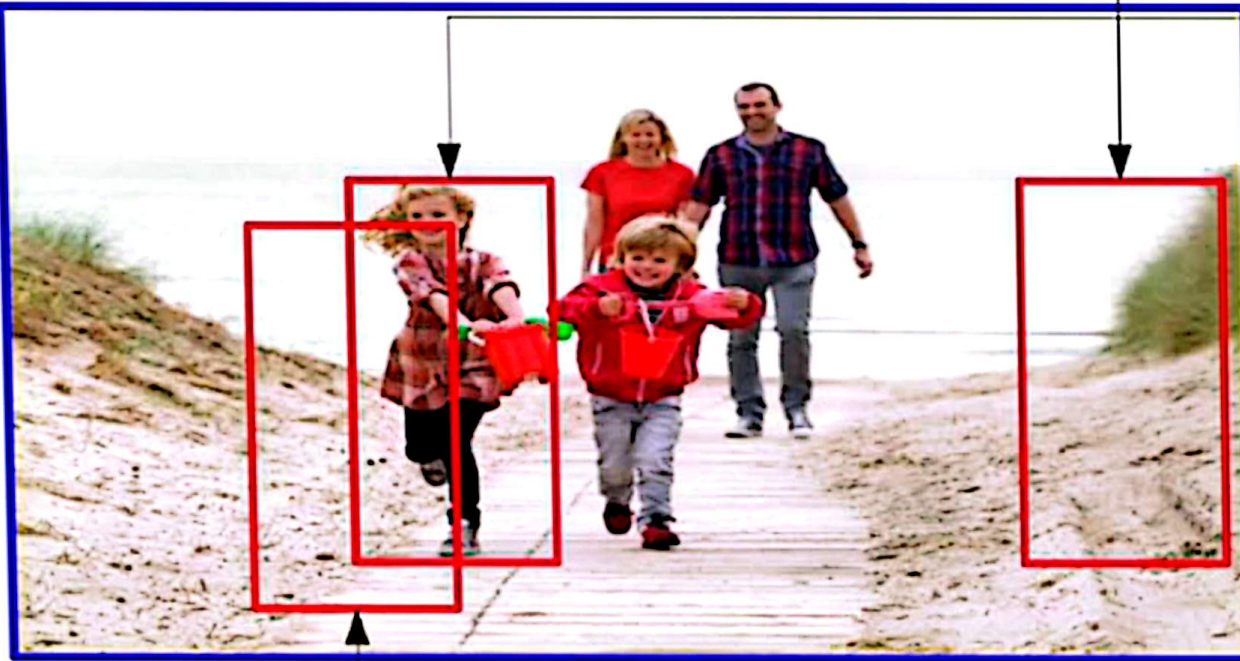
Detection + Localization



Different Windows Sizes
Scan Multiple Times



Sliding Window Technique



Detect Three Classes

C_1 Person

C_2 Cat

C_3 Book

Final Filter is $1 \times 1 \times 8$

(to detect 3 classes)

1
0.3
0.1
0.6
0.9
1
0
0

0

X
X
X
X
X
X
X
X

0.7
0.2
0.2
0.6
0.9
1
0
0

P
B^c
B^x
B^y
B^w
B^h
C₁
C₂
C₃

Sliding Window Technique

- **Very Very Slow (Computational Cost)**
- **Pass through the Image Many Times with Different Sizes**
- **If Stride is large, (loss of precision in localization)**